

The Dispatcher System

- [Setting up the System to Dispatch a Train](#)
 - [Introduction](#)
 - [Setting up the System - Summary](#)
 - [Stage 1: Prerequisites](#)
 - [Stage 2: Setting up the System](#)
 - [Prerequisites - Details](#)
 - [Setting Up the Dispatcher System - Details](#)
 - [Running Stage1: Editing the panel and producing the transits](#)
 - [Running Stage2: Final Checks](#)
 - [Starting the Program](#)
 - [Starting the Program](#)
 - [Running the System](#)
 - [Setup or Register a Train](#)
 - [Setup two or more trains at once](#)
 - [Saving and Retrieving Results from File](#)
 - [Modify existing trains](#)
 - [Dispatch or run a Train Directly](#)
 - [Example: Run a train between two stops \(stations\)](#)
- [Running Trains along Routes](#)
 - [Setup a Route](#)
 - [Actions](#)
 - [View/Edit Routes](#)
 - [View Routes](#)
 - [Editing the route](#)
 - [Run Route](#)
 - [Run One Route](#)
 - [Run Several Routes at Once](#)
- [Running Trains to Schedules](#)
 - [Introduction](#)
 - [Setting Up Schedules](#)
 - [Adding Schedules](#)
 - [To save the set of scheduled trains](#)
 - [To retrieve a set of scheduled trains](#)
 - [Setting Departure Times](#)
 - [Setting Departure Times Manually](#)
 - [Setting Departure Times Automatically](#)
 - [Set Wait Time in Stations](#)
 - [Run Train on Route to Set Journey Times](#)
 - [Run Scheduled Trains](#)
 - [Initialisation: Change Running Period Parameters](#)
 - [Start the Scheduled Trains](#)
 - [· Set the time to a period within the running period](#)
 - [· Start the scheduling of the trains](#)
 - [· Start the clock](#)
 - [Stop Scheduling Trains](#)
- [Viewing the Times of Scheduled Trains on Timetables](#)
- [Introduction](#)
- [Set Platforms and Station Groups](#)
- [Timetables viewed on Computer running JMRI](#)
- [Turning Timetables Off](#)
- [Timetables viewed on Tablets / other Computers on same local Network](#)
 - [Introduction](#)
 - [Generating the MQTT Node Red code](#)
 - [Setting up the required Node Red environment](#)
 - [Setup a MQTT Connection in JMRI](#)
 - [Display the MQTT timetable](#)
- [Modifying the Behaviour of Running Trains](#)
 - [Set Stopping Distance](#)
 - [Modify All Stopping Distances](#)
 - [Modify Stopping Distance on one Transit](#)
 - [Set Stop Sensor](#)
 - [Operation of Stop Sensors](#)
 - [Set Station Wait Time](#)
 - [Modify all station wait times](#)
 - [Modify the wait time at a particular station approach](#)
 - [Set Station Direction](#)
 - [Setting Station Direction at a Station](#)
 - [Setting the Station Direction using a block](#)
 - [Restrict Transit Operation](#)
 - [Introduction](#)
 - [Creating the required Rule](#)
- [Keeping a Scheduled System Running](#)
 - [Moving the Train](#)
 - [Changing the Direction of the Train](#)
- [Adding New Stations or Track to a Completed Panel](#)

Setting up the System to Dispatch a Train

Introduction

The Dispatcher System is designed to enable you to run trains between locations (normally sidings or stations, but really any set of blocks) designated by yourself.

Provided you have set up a system as described in System Requirements, by pressing a few buttons on the System Generation Panel, a complete system will be set up for you complete with Transits, Train Info files, and buttons for you to press which will route trains across your layout.

A network will be set up behind the scenes which will find the best route from the train's location to the required location and set in place a set of transits which will guide the train to the required location.

The train will either stop at all stations on the route, or travel without stopping at intermediate stops, depending on whether express or stopping mode is selected.

Provision is made for multiple train operation, the trains stopping for each other. This is a feature of Dispatcher, but the system allows easy selection of each train to control each one.

Setting up the System - Summary

There are two stages to set up the system for automatic train running:

Stage 1: Prerequisites

Fulfil the prerequisites, which include having speed profiled your locos that you wish to run with the system

- Start JMRI with a panel loaded.
- The panel should have:
 - SignalMasts. Do not use Signal Heads. This is an alternative way of setting up signals and Dispatcher system will not work with them.
 - Signal Masts should be placed on all potential section boundaries.
 - No Sections. Sections will be created using the signals, and any existing sections will be deleted.
 - No SignalMast Logic (if there is any they will be deleted and replaced)
 - No Transits (if there are any they will be deleted and replaced)
 - No Train Info Files (if there are any they will be deleted and replaced)

In other words there is a lot of stuff that you would ordinarily need to set up to use Dispatcher, that you don't need to using Dispatcher System

In addition the block table should have been edited to indicate the blocks where the train should stop

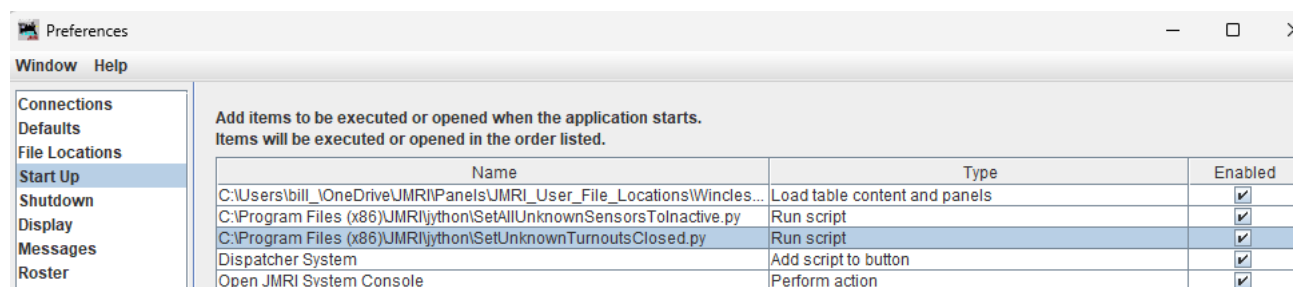
These requirements are explained in detail below.

You will be using Sensors and Signal Masts. In order for these to work properly, people usually set up a start of day operation. As a quick way of getting the system working you can put two routines in the startup file in properties, which can be found in the scripts area

- SetAllTurnoutsClosed.py and
- SetAllUnknownSensorsToInactive

If you use these be sure to load the panel before calling these in the startup routines, else they will not work. It is also useful to put a button on the startup screen calling the dispatcher system file in the same directory

- DispatcherSystem.py



Stage 2: Setting up the System

To run Dispatcher System:

- Bring up the Dispatcher System Panel
- Run Stage1 to generate everything to run Dispatcher between stations
- Run Stage2 to check the dispatcher options
- Run the trains using the commands on a newly produced command panel

Prerequisites - Details

To use the Dispatcher System there are several prerequisites

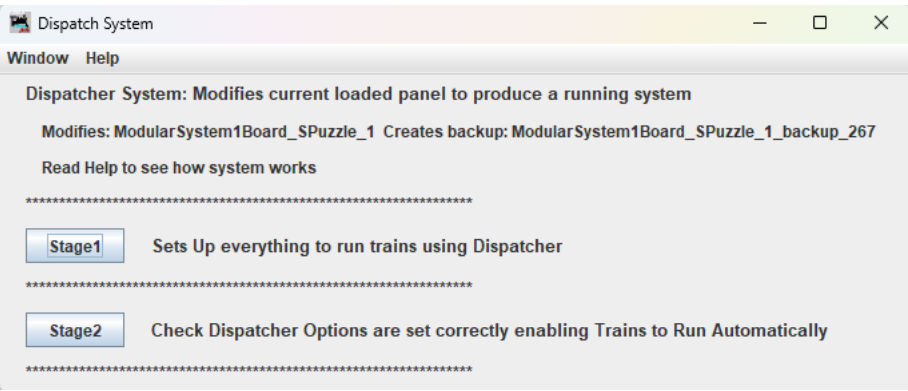
- You should have set up the system with signal masts.
- The locos you want to run on the system should be:
 - entered in the Roster
 - set up with a speed profile
- the required stopping places (blocks) should be indicated by entering stop in the comments field for the appropriate block
- Provision is made for a station announcement to be made when a train enters a station.
- If the block name is unpronounceable, a good stop name can be entered in the comments field for the departure announcements
- Typical blocks comments would be

Blocks								
File View Paths Speeds Values Window Help								
Turnouts	System ...	User Name ▲	Value	Comment		Edit	Direction	Length
Sensors	IB:AUTO...	CornerDown		stop "Corner Down"	Delete	Edit	None	39.37
Lights	IB:AUTO...	CornerUp		stop "Corner Up"	Delete	Edit	None	47.24
Signal Heads	IB:AUTO...	DownLineExit			Delete	Edit	None	4.84
Signal Masts	IB:AUTO...	DownLinePlatform		stop	Delete	Edit	None	33.46
Signal Groups	IB:AUTO...	EntryTo Parcels			Delete	Edit	None	4.72
Signal Mast Logic	IB:AUTO...	FY7			Delete	Edit	None	10.00
Reporters	IB:AUTO...	FY8			Delete	Edit	None	10.00
Memory Variables	IB:AUTO...	HeadShunt		stop	Delete	Edit	None	15.75
Routes	IB:AUTO...	ML-MLD		stop	Delete	Edit	None	15.75
LRoutes	IB:AUTO...	ML-MLU		stop	Delete	Edit	None	15.00
Logix	IB:AUTO...	ParcelsSiding		stop	Delete	Edit	West	19.69
LogixNGs	IB:AUTO...	SP-GAP			Delete	Edit	None	3.15
LogixNG Modules								
LogixNG Tables								
LogixNG Global Variable								
Blocks								
Sections								

Setting Up the Dispatcher System - Details

Running Stage1: Editing the panel and producing the transits

Run DispatcherSystem.py in the main jython directory. You will get something like this:



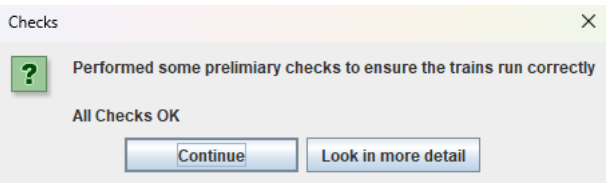
You are prompted to run Stage1 which adds all the icons required to run the system on a new pane, and generates transits

If the old panel name was WR.xml, the new panel will be backed up to WR_backup_xx.xml where _xx is an incremental number which is added if previous backups exist

Some checks are done first, we check that:

- All blocks have a sensor
- No two blocks have the same sensor
- stops are set up
- all blocks have lengths

If all checks pass we get a message:



Further details of the checks can be viewed if required. If the checks do not pass details of the errors are given.

If no trains are present in the roster with speed profiles, this will be picked up at run time, and can be corrected then.

The following items are then produced automatically, with countdown timer showing the progression

- A backup of the existing panel
- The changes to the panel with the stopping points and stopping sensors
- The signal logic to operate the signals
- The required sections
- The transits and train info files required to move trains between stopping points

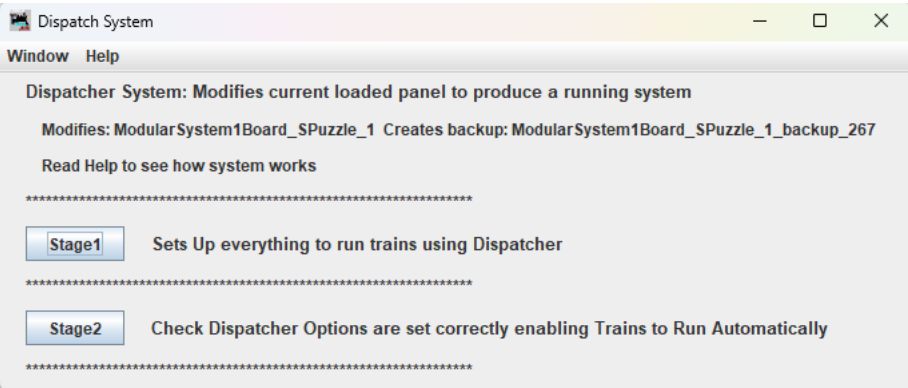
All Sections, Transits and TrainInfo Files produced.

The JMRI tables and panels have been updated to support the Dispatcher System
A store is recommended.

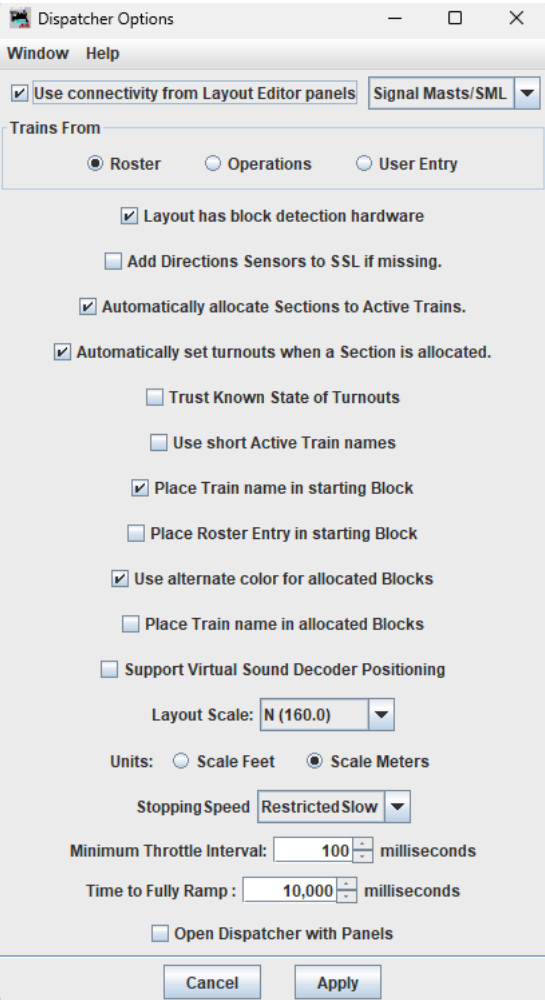
OK

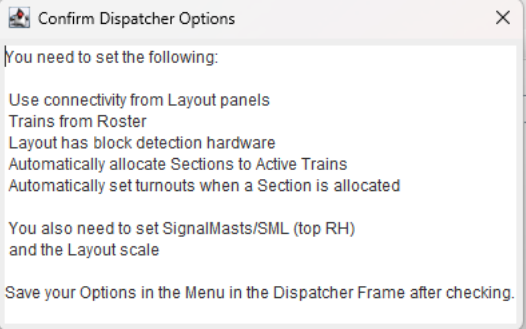
It is recommended that the panel is saved after testing. The current filename can be used if desired as a backup has been done automatically.

Running Stage2: Final Checks

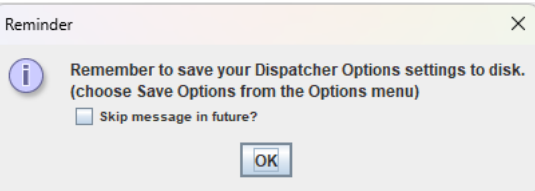


Click Stage2 and one gets two screens, one opening the Dispatcher options screen and another saying the options that need to be set for the Dispatcher system to work

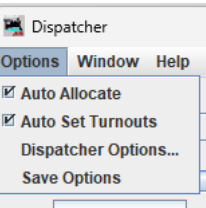




Set the required option and we get a message reminding you to save the changed options



Select the dispatcher window that has opened automatically and click => Save Options

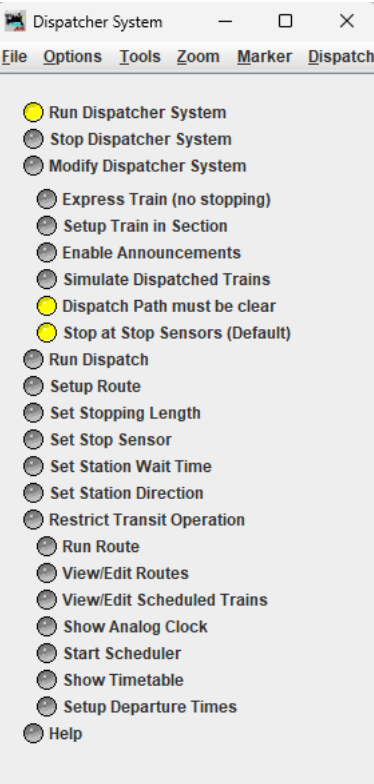


Starting the Program

The program needs to be started. The program is a Jython script, which is started from the Dispatcher System Window which has been produced in the previous stages.

Starting the Program

You will notice several buttons in the top RH corner.



The first two (run and stop dispatcher system) start and end the threads required for the program to run. The threads need to be in operation for the other buttons to work.

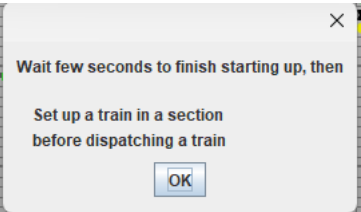
While operating the dispatcher system, if you have the thread monitor running (Panels => Thread Monitor) you can monitor what processes are running.

The thread monitor will look like the following once the system is started

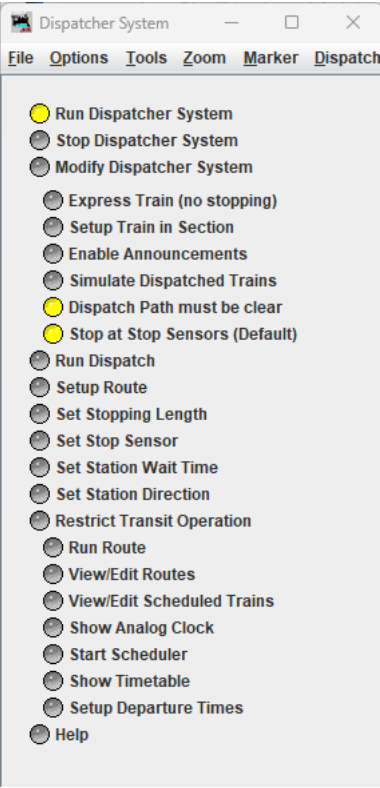
Thread Monitor

Window Help			
Name	Cycles	Kill	
New Train Master	0	...	
Stop Master	0	...	
Reset Button Master	0	...	
Dispatch Master	0	...	
Simulation Master	0	...	
Scheduler Master	0	...	
Monitor Track Master	16	...	
Off-Action Master	0	...	

Press the run button. The following screen appears



Press OK, and wait for the Express Train and the Run Dispatch options to be highlighted and the run Dispatch option will cause the following prompt to be displayed:



Running the System

To run the system the following needs to be set up

1. Place a train on the layout. If you wish to simulate the train to get an idea of how the system will operate click the bottom left button of a station which will make the station occupied.
2. **Register the train** with the dispatcher system telling the system what way it's facing so the system knows what transit to use to move the train (forwards or backwards). Press the 'Setup Train in Section' button to do this.
3. **Run the train.** This can be done in several ways.
 1. **Setup a Dispatch.** Click the Run Dispatch Button and click a station button and specify which registered train will move towards that station
 2. **Setup a Route** and run the route. A Route consists of a path joining several stations. The stations joined can be anywhere on the layout no matter how far apart they are, and the system will find valid paths between the stations. The station buttons can be used to either set up a dispatch or set up a route depending upon whether "Setup Route or "Run Dispatch" is selected. The route is run by pressing the "Run Route" button.
 3. Setup routes, Setup Scheduled Trains and then start the scheduler.
4. If a physical train is on the layout at the start of the dispatch the train should move. If you have just made the block occupied with no physical train you can press the 'Simulate the Dispatched Train' button to get an idea of how the train will run. The train will simulate much quicker than a real train and takes no account of the train length.
5. Announcements can be activated by pressing the 'Enable Announcements' button. This is off by default as they can get a but irritating.

We go over each of these stages in detail below:

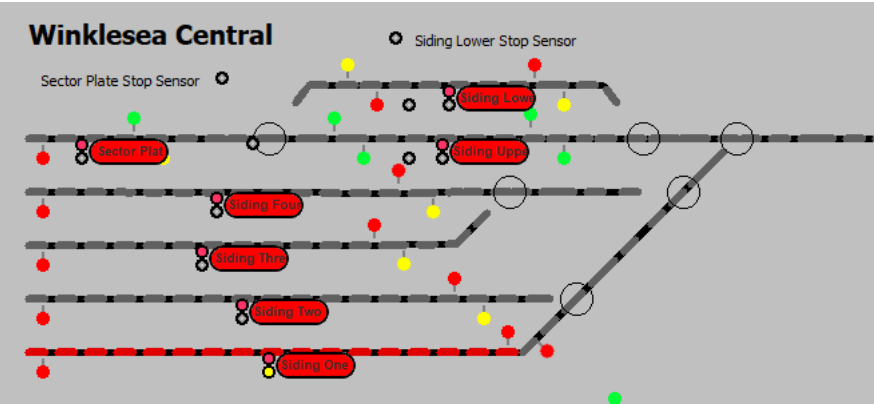
Setup or Register a Train

The first step is to register a train in a siding or at a stop. If you are simulating the system to see whether it will work before running trains you may press the bottom left hand small button which will make the section occupied. Drive a train to the required stop, and press Setup Train in Siding.

We need to tell the system which way the train is pointing.

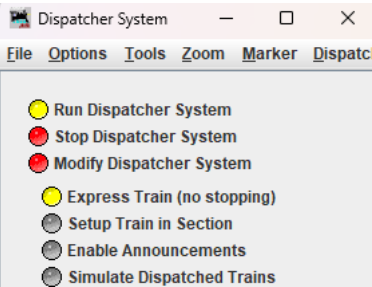
- At a siding it's easy there is one direction the train can go, and we say which way it is facing out of the siding
- At another stop we have to indicate which way we are talking about and we highlight one direction

suppose we have the following layout with a train in Siding One

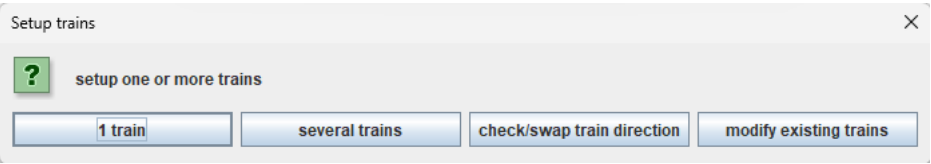


Choosing the Train

We click “Setup Train in Section”

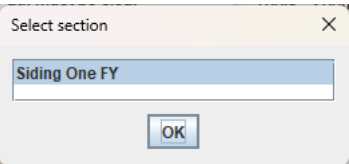


If the train is Siding One as indicated one gets the following message

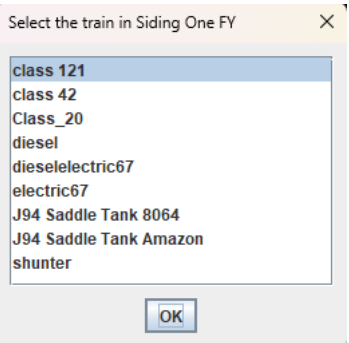


A facility exists to set up several trains at once. Assume for now we are setting up just one train. Click the 1 train button.

A dropdown with the blocks containing trains that have not been assigned. Choose the block you wish to assign a train to.



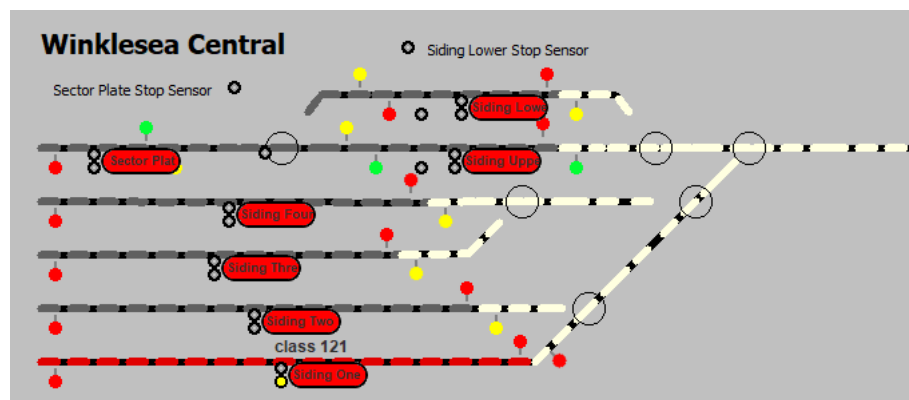
You get a list of unassigned trains with speed profiles.



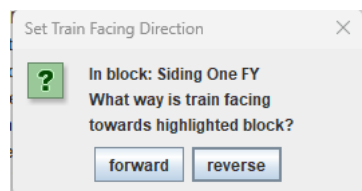
Select the train that is in the block selected (in our case Siding One)

Register the Direction the train is facing.

A block next to the selected block will be highlighted:



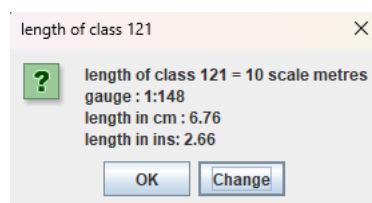
The following message then pops up



Select the correct direction and click OK.

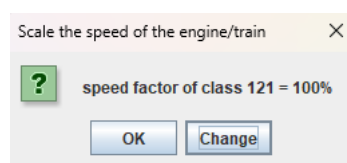
The train will appear in the memory location for the block.

A message giving the default length of the train will appear in scale metres and the equivalent for the scale chosen in cm and inches. Note the scale is set in Preferences:Warrants:



Set the length of the train to the actual length. This will be used to stop the train accurately.

A message giving the speed factor of the train. If the train runs too slowly increase the speed factor, if it runs too fast, decrease it. Normally it will not have an effect if you increase by more than 20%.

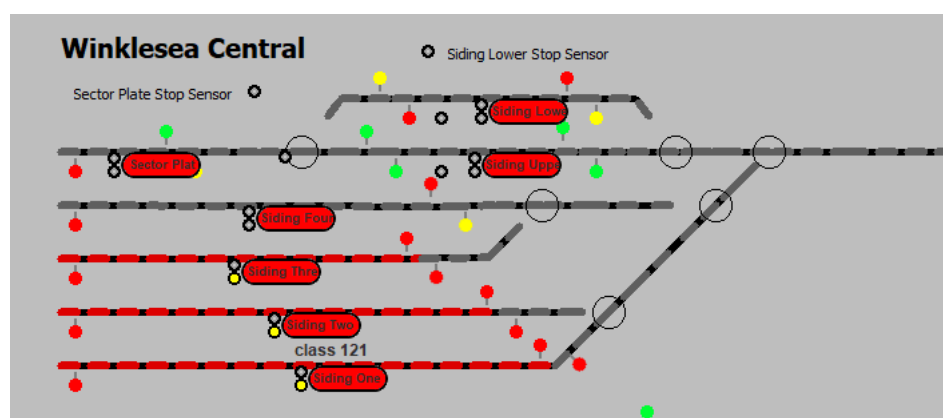


Setup two or more trains at once

If you wish to setup two or more trains at once, or if you always setup certain trains at a given starting position, an alternative way of proceeding is provided.

Suppose we wish to setup trains at Siding 2 and Siding 3.

Position trains in the blocks (or of simulation click the occupancy buttons to the bottom left of the red buttons).



Click Setup Trains as before:

Setup trains

?

setup one or more trains

1 train

several trains

check/swap train direction

modify existing trains

Click several trains. The following screen appears:

Set up trains. Any trains shown in blocks are showing on the track but not allocated. Save to allocate

Train	Block	Set Direction	Direction Facing	Length (scale metre...	Speed Factor
	Siding One FY	click ->	☐	10	100
	Siding Two FY	click ->	☐	10	100

Save

Close

Populate

Save To File

Load From File

Delete Saved Files

Rows appear for the two blocks that have non-allocated trains.

Select the trains from the dropdown menus

Click -> means click the direction Facing selection box in the next column. When this is clicked, the next block to the occupied block is highlighted as before and the directions selected populate the set Direction column. The Length and Speed Factor can be changed by editing the values given.

Set up trains. Any trains shown in blocks are showing on the track but not allocated. Save to allocate

Train	Block	Set Direction	Direction Facing	Length (scale metre...	Speed Factor
J94 Saddle Tank 8064	Siding One FY	forward	☐	10	100
class 121	Siding Two FY	forward	☐	10	100

Save

Close

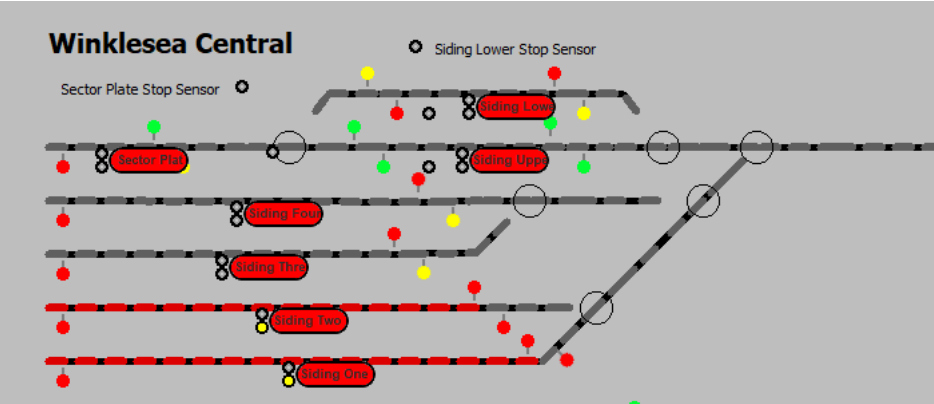
Populate

Save To File

Load From File

Delete Saved Files

Click Save, and the trains will be shown in the panel.



Saving and Retrieving Results from File

It gets a bit tedious entering the details for trains a lot, so the results can be saved to file, and retrieved. Before saving click the Save to File button.

Select a .txt file

Save In: setup_trains

File Name: sidings 1 and 2

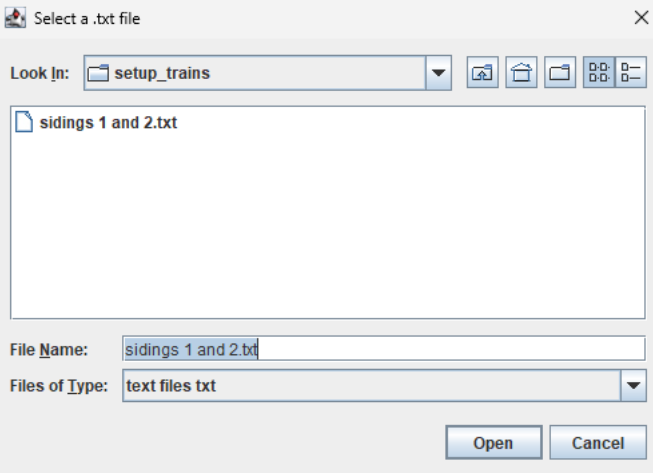
Files of Type: text files txt

Save

Cancel

Enter the required name (no need to put the .txt suffix on) and press save.

To load the saved file click the load button, and if the folder gets messy use the delete saved file option.



Modify existing trains

This option allows trains to be modified or at least viewed that have already been setup. Press the button, and modify the trains as required. Previously setup trains can be saved as per the previous step.

Dispatch or run a Train Directly

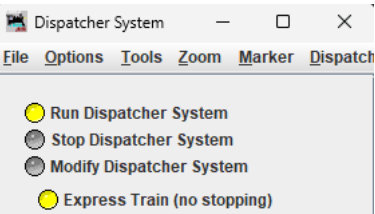
To set up a dispatch a train must have been registered with the system, as described in the previous section, and located in a stop.

Click the Run Dispatcher System and the following prompt appears.



Click OK.

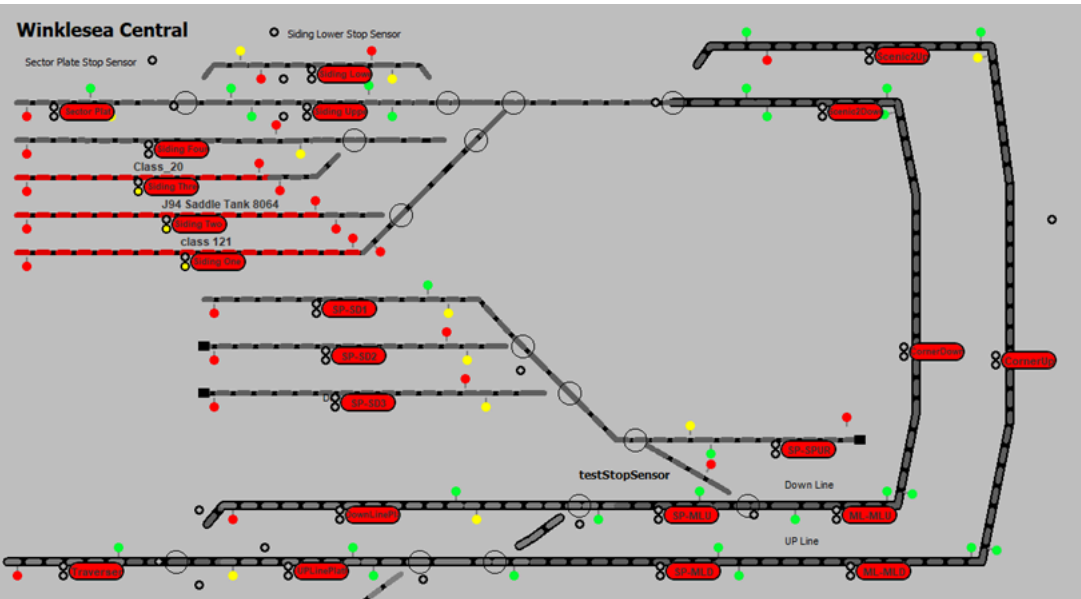
Clicking the red station buttons when the Run Dispatch option is selected will allow a train to run between any two stations. The shortest route (using the lengths of the blocks) will be found. If the route requires the train to reverse then the system will cope with this. There are two options allowed Express where the train stops at the minimum stations possible or stopping when the train stops at all stations en-route. We will be running the train in express mode.



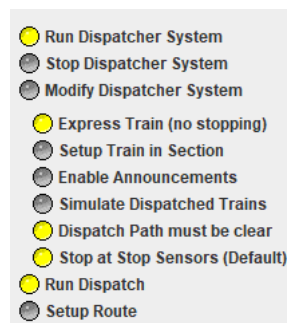
If you wish to run in stopping mode deselect the above button.

Example: Run a train between two stops (stations)

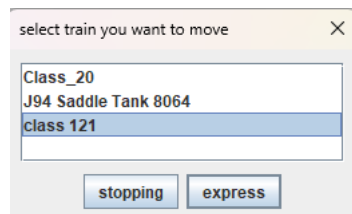
Suppose one has the following layout with a train in siding1. Suppose we wish to move the train to Upline Platform (near the bottom left of the panel).



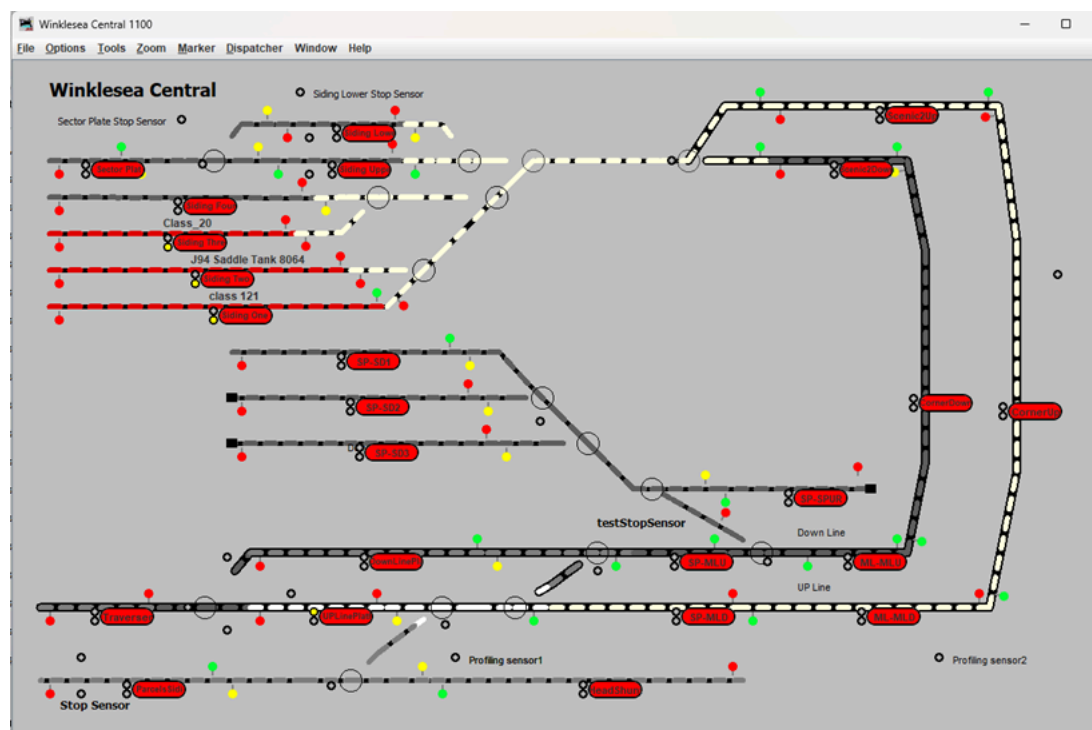
To move the train to a station, click the destination station, ensuring first that the option Run Dispatch is selected (this option is selected at startup, but may be deselected if other operations have been utilised).



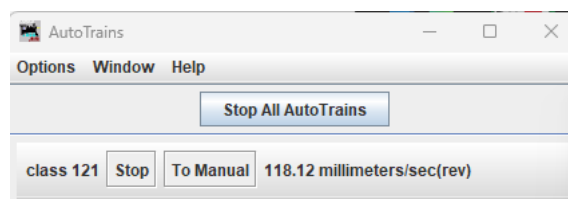
We wish to move train class 121 from Siding 1 to UpLinePlatform. Click the UplinePlatform button. A prompt appears with the trains registered on the system. Select the one you wish to move and press the mode you wish the train to move in express (stopping in the minimum number of stops) or stopping (stopping at all stations en-route).



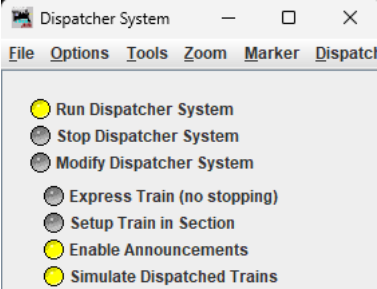
Select the required train from the drop down, and a dispatch will be set up to the required stop (UpLinePlatform).



The train will start moving, and the speed indicated in the Auto trains window.



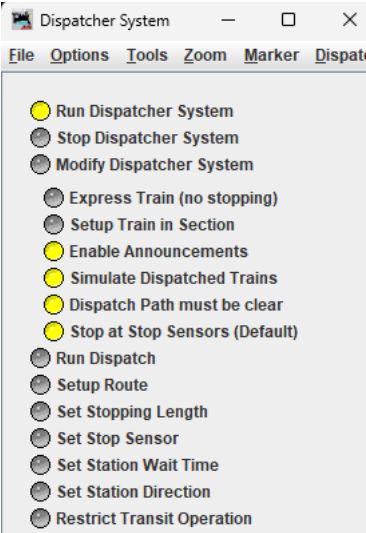
If you are simulating the movement of the train, then the Simulate Dispatched Trains button must be selected before the simulated movement starts.



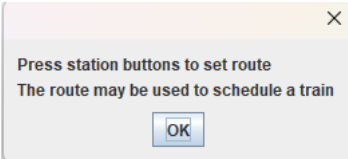
Running Trains along Routes

If you wish to do something more complicated that stop at no stations or all, then routes are useful. With routes you can specify several places that the train will stop off at en-route to the destination, and not just all or none.

Setup a Route



Click 'Setup Route'. Run Dispatch will be deselected, and Setup Route Selected. The following message will be displayed:

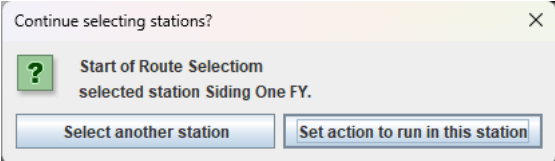


Suppose we wish to send a train from Siding One to Upline Platform stopping at CornerUp

We need to select the following stations in order

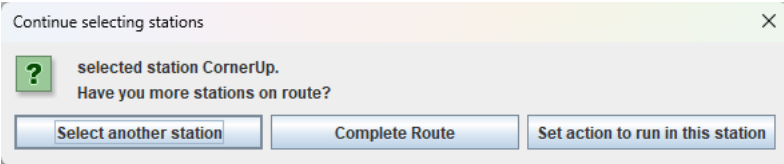
- Siding One
- Corner Up
- Up Line Platform

Select Siding One. The following popup will be displayed

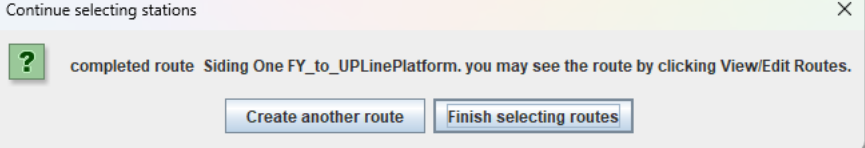


Actions or special scripts can be run at stations of a route. We will describe these below. Select ‘Select another station, and press Corner Up station

We get the following:



Press Select another station, click UpLinePlatform, then press Complete Route. The route is created and confirmation is acknowledged.

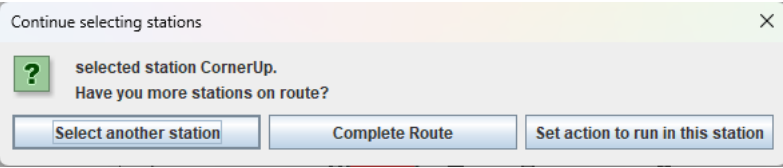


Another route may be created, or here we press Finish selecting routes. The setup Route button is deselected. To run a dispatch or setup another route the appropriate button must be selected.

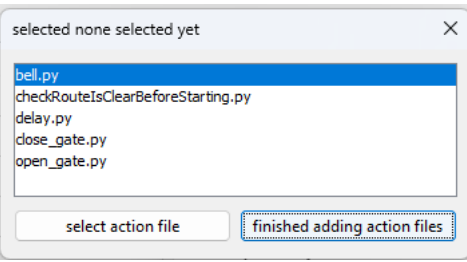
Actions

Actions are python files which may be run at stations. A (very) few action files are provided in the Dispatcher System, and other files may be added. The provided files are in the jython/DispatcherSystem/actions folder in JMRI. It is best not to add new files there since they will be wiped out when a update is installed. Instead use them as a guide, and add them in the userfiles directory (see directories in preferences) in the folder dispatcher/actions. If the actions folder does not exist, create it.

Suppose we are selecting stations when setting up a route and get the following menu:

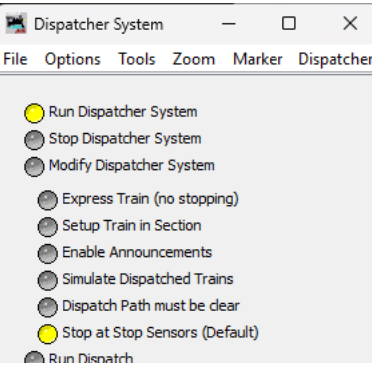


Click 'Set Action to run in this station' and you will get something like the following:

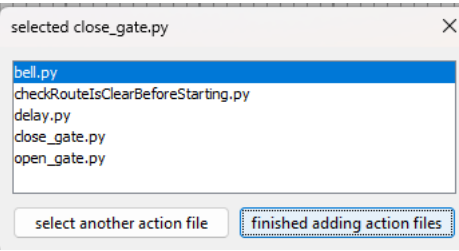


Where the first 3 actions are the actions that are provided in the dispatcher system (at the time of writing this document) and open and close gate are actions that have been added to my user directory. View the python of the 3 actions to see how to create your own files.

Bell.py rings a bell, checkRouteIsClearBeforeStarting.py allows another way to cause the train to wait if the global Dispatch Path must be clear is not selected. Delay delays for 10 seconds which can be changed by editing the file.



If we select close_gate.py we get the following:



And we can finish adding actions files and complete the route.

We can examine the route by selecting View/Edit Routes, selecting the route and clicking edit. A typical route with actions might be:

Train Route: block4_to_block1_1

Station / Action	Duration (secs)	Duration (f mins)	Departure Time	Delete Row	Use of Stop Sensor
block4		0	00:00	☐	
open_gate.py				☐	
block1	0	0	00:00	☐	Use Default
dose_gate.py				☐	

Close

Toggle Journey/Wait Times

Calculates to the precision of 1 fast minute. 6 seconds = 1 fast minute

View/Edit Routes

View Routes

To see the route that has been created press View/Edit Routes

- Run Dispatch
- Setup Route
- Set Stopping Length
- Set Stop Sensor
- Set Station Wait Time
- Set Station Direction
- Restrict Transit Operation
- Run Route
- View/Edit Routes

You will see something like this:

Routes

Route	Edit	Delete Row
CornerUp_to_Siding Upper FY	☐	☐
Siding One FY_to_UPLinePlatform	☐	☐
SP-MLD_to_ParcelsSiding	☐	☐
SP-MLU_to_ML-MLU	☐	☐
SP-SD1_to_SP-SD1	☐	☐
SP-SD3_to_SP-SD1	☐	☐
SP-SPUR_to_SP-SPUR	☐	☐

Close

Toggle Scheduled Routes

Delete All Rows

If you press Toggle Scheduled Routes you will see only the scheduled routes. Suppose we had two scheduled routes (see later for description) you would get the following when pressing that button. If you have no scheduled routes then you will get a blank window.

Scheduled Routes

Route	Edit	Delete Row
SP-MLD_to_ParcelsSiding	☐	☐
SP-SPUR_to_SP-SPUR	☐	☐

Close

Toggle Scheduled Routes

Delete All Rows

Click the edit button in the row of Siding One FY_to_UplinePlatform and you see:

Train Route: Siding One FY_to_UPLinePlatform

Station / Action	Duration (secs)	Duration (f mins)	Departure Time	Delete Row
Siding One FY		0	00:00	☐
CornerUp	0.0	0	00:00	☐
UPLinePlatform	0.0	0	00:00	☐

Close

The amount of time spent at each station (Duration) is seconds and fast minutes are needed to calculate the Departure time required for scheduling trains. These are not needed at this stage so we ignore them.

· For now, note that the route stops at the stations we requested, Siding One. Corner Up and Up Line Platform.

Editing the route

If you wish to delete a station from a route you may select the route as shown in the previous section, and press the delete row check box for the station you want to delete.

For example clicking the delete box for CornerUp will give you:

Train Route: Siding One FY_to_UPLinePlatform

Station / Action	Duration (secs)	Duration (f mins)	Departure Time	Delete Row
Siding One FY		0	00:00	☐
UPLinePlatform	0.0	0	00:00	☐

Close

Which does not stop at CornerUp.

If you wish to add a station, no facility to do this is given within Dispatcher System per se, and the advised action is to build another route from scratch.

Note: For Experts: Dispatcher System uses Operations Pro to create routes, and the routes can be edited more fully within Operations Pro: Routes

Run Route

To run a route a route must already have been set up. Click Run Route

Dispatcher System

FileOptionsToolsZoomMarkerDis

Run Dispatcher System

Stop Dispatcher System

Modify Dispatcher System

Express Train (no stopping)

Setup Train in Section

Enable Announcements

Simulate Dispatched Trains

Dispatch Path must be clear

Stop at Stop Sensors (Default)

Run Dispatch

Setup Route

Set Stopping Length

Set Stop Sensor

Set Station Wait Time

Set Station Direction

Restrict Transit Operation

Run Route

?

run one or more routes

1 route

several routes

You will be prompted to run one or several routes at once.

Run One Route

Select 1 route, You are prompted with the train to move. If the train is not at the start of the route it will move to the start before running the rout.

You will be prompted with all the trains you have set up

what train do you want to move?

Class_20

J94 Saddle Tank 8064

class 121

OK

Choose the train class 121, a list of routes are displayed.

choose route

Siding One FY_to_UPLinePlatform

SP-MLD_to_ParcelSiding

SP-MLU_to_ML-MLU

SP-SD1_to_SP-SD1

SP-SD3_to_SP-SD1

SP-SPUR_to_SP-SPUR

OK

show routes starting at train position

Select show routes stating at train position.

choose route

Siding One FY_to_UPLinePlatform

OK

show all routes

We see that train class 121 is at the starting position of route Siding One FY_to_UplinePlatform.

Select OK.

What do you want to do

stop at end of route

return to start position

return to start and repeat

cancel

OK

Select whether you want to

- Run the route and stop
- Run the route and return to the start position (not necessarily along the same route, but the shortest route)
- Run the route, return to the start position and repeat

We will select stop at end of route

The train route is setup and the train starts moving along the route.

Run Several Routes at Once

Click Run Route. You are prompted for

?

run one or more routes

1 route

several routes

Click several routes

Allocate Routes

Train

Route

Run Route

Task

Delay (secs)

No. Repetitons

Class_20

☐

stop at end of route

10

3

J94 Saddle Tank 8064

☐

stop at end of route

10

3

class 121

☐

stop at end of route

10

3

Populate

Tidy

Run Routes

Close

Task

Delay

No. repetitions

Save To File

Load From File

The above panel appears populated with a row for each train that has been setup in the previous section. Allocate routes to each of the trains that you wish to move. The default is for the train to wait 10 seconds before moving the train. Change this as required. If the task chosen for a train is to make several repetitions, the no of repetitions in the last column will be used. We may specify for example

Allocate Routes

Train	Route	Run Route	Task	Delay (secs)	No. Repetitions
Class_20	Siding One FY_to_UPLinePlatform	☐	stop at end of route	10	3
J94 Saddle Tank 8064	SP-MLD_to_Parcels Siding	☐	return to start position	5	3
class 121	SP-SPUR_to_SP-SPUR	☐	return to start and repeat	15	3

PopulateTidyRun RoutesCloseTaskDelayNo. repetitionsSave To FileLoad From File

Note that the trains will move in the order J94, Class_20, Class 121, and the route SP-SPUR_to_SP-Spur will repeat 3 times.

Note that Class_20 will run its route Starting from Siding One, whereas J94 will move to SP-MLD first then run the route from SP-MLD to Parcels Siding, similarly Class 121 will travel to SP-SPUR before running the route.

We will go over both options: ‘Several routes’ is used to run several routes setting off at the same time.

Running Trains to Schedules

Introduction

Trains can be scheduled to run at specific times over routes. In the schedule table the trains on the routes are set to run at specific start times. They can then be set to run just at that time or at intervals based on that start time, say every 20 minutes.

The running period per day can be set from say 4am to 11pm, and the fast clock set to a rate within that period, and optionally at a faster rate outside that period (during the night).

Timetables can be set up to display the schedules trains, For timetables to work the start time of each station on each route that is scheduled needs to be set up. This can be done manually or automatically, the latter by running a train over he route using a special routine.

Timetables can be displayed locally on the computer running JMRI, or on remote computers/tablets communicating via MQTT usually over a local network.

The schedules are set up from the Dispatcher panel. At the bottom of the panel you see the following screen.

Run Route

View/Edit Routes

View/Edit Scheduled Trains

Set Scheduler Start Time

Show Analog Clock

Start Scheduler

Show Timetable

Setup Departure Times

Help

Click “View/Edit Scheduled Trains” and a screen pops up similar to the following:

Scheduled Trains

Time	Route	Repeat	Don't Schedule	Train Description	Edit_Row	Delete Row
------	-------	--------	----------------	-------------------	----------	------------

AddSave & CloseRepeatDelete All RowsSave To FileLoad From FileDelete Saved Files

Setting Up Schedules

Adding Schedules

Press the Add button. A row is added to the screen

Scheduled Trains

Time	Route	Repeat	Don't Schedule	Train Description	Edit_Row	Delete Row
00:00		Once	☐	Select Train	☐	☐

AddSave & CloseRepeatDelete All RowsSave To FileLoad From FileDelete Saved Files

Press the dropdown for route

Scheduled Trains

Time	Route	Repeat	Don't Schedule	Train Description	Edit_Row	Delete Row
00:00	Siding One FY_to_UPLinePlatform	Once	☐	Select Train	☐	☐

Add

DownLinePlatform_to_Siding One FY
ParcelsSiding_to_Siding Two FY
ParcelsSiding_to_Siding Two FY_1
Siding One FY_to_UPLinePlatform
Siding Two FY_to_ParcelsSiding
SP-MLU_to_DownLinePlatform
SP-MLU_to_ML-MLU
SP-SD1_to_SP-SD1

Delete All Rows Save To File Load From File Delete Saved Files

Select one of the routes that have been set up. We select Siding One FY to UpLinePlatform.

Press the dropdown for Repeat

Scheduled Trains

Time	Route	Repeat	Don't Schedule	Train Description	Edit_Row	Delete Row
00:00	Siding One FY_to_UPLinePlatform	Repeat every 30 mins	☐	Select Train	☐	☐

Add Save & Close Repeat

Once
Repeat every 20 mins
Repeat every 30 mins
Repeat every Hour
Repeat every 2 Hours

Delete All Rows Save To File Load From File Delete Saved Files

We have the options of running just at the given time, or repeat at given frequencies, Once, Every 20 mins, Every 30mins, Every Hour, or Every 2 hours. We would like the train to repeat every 30 mins, so we select that.

Set the time. The hour is ignored if we are running at a set frequency, so we can leave it at 00, (note if we have set to Once, the hour is used). We set the minutes to say 2 minutes past the hour so the train will run a 2 and 32 minutes past the hour. Double click the time to set.

Scheduled Trains

Time	Route	Repeat	Don't Schedule	Train Description	Edit_Row	Delete Row
00:02	Siding One FY_to_UPLinePlatform	Repeat every 30 mins	☐	Select Train	☐	☐

Add Save & Close Repeat

Delete All Rows Save To File Load From File Delete Saved Files

Select the train that will be running on the schedule

Scheduled Trains

Time	Route	Repeat	Don't Schedule	Train Description	Edit_Row	Delete Row
00:02	Siding One FY_to_UPLinePlatform	Repeat every 30 mins	☐	class 121	☐	☐

Add Save & Close Repeat

Delete All Rows Save To File Load From File Delete Saved Files

class 121
class 42
Class_20
diesel
dieselectric67
electric67
J94 Saddle Tank 8064
J94 Saddle Tank Amazon

Edit Route displays the stations on the route, and allows us to enter dispatch times, which will be explained later.

Delete Row deletes the row without warning, so if you have several trains dispatched, it may be best to save these to file to avoid having to type them in again.

Delete all Rows does what it says.

The Repeat button toggles all the entries through Once, Every 20 mins, Every 30mins, Every Hour, or Every 2 hours

Save to File saves the entries to file, Load from File allows you to load previously saved schedules, and wipes out existing ones

Delete Saved Files enables you to only have saved the files that you want.

Saving Schedules

Click the save button to save the schedule. Only valid complete rows are saved.

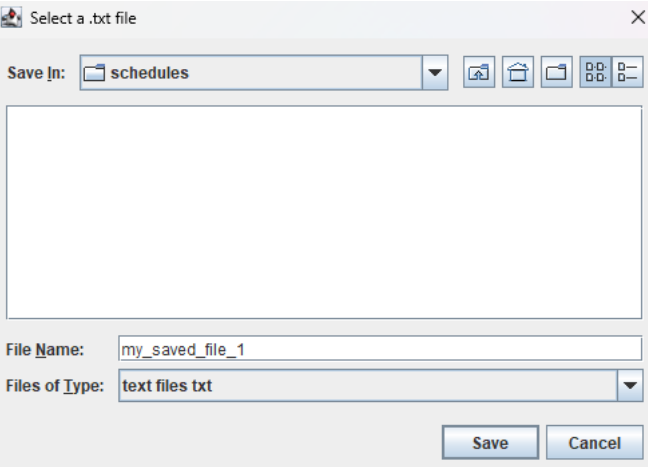
To save the set of scheduled trains

Press save to file, and select an appropriate file name

Saving Valid rows

OK

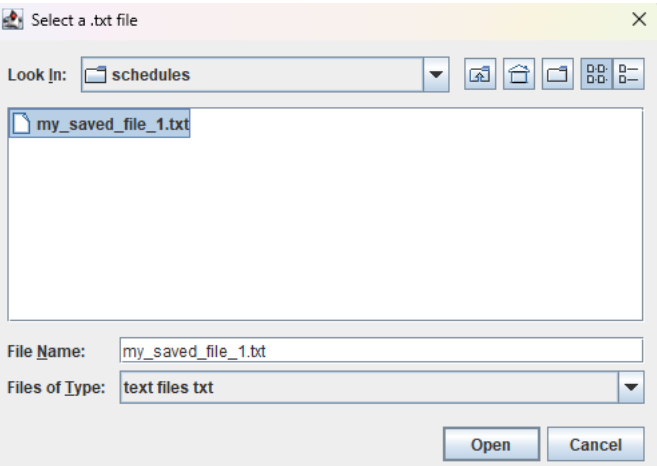
Only complete valid rows are saved



Save to my_saved_file_1

To retrieve a set of scheduled trains

Press Load from File. Beware this will overwrite the existing schedules. It may be advisable to save the existing schedules to file before doing this.



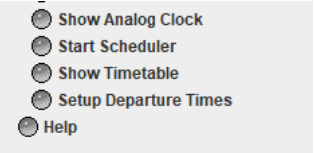
Click the required file (here there is only one). The saved file will be loaded.

Setting Departure Times

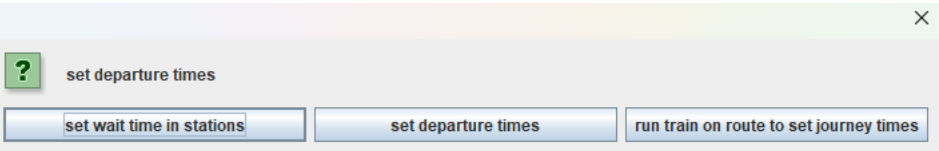
If departure times are set for stations on a route, then the trains will wait till the departure time, before setting off. Also the departure times will be shown on any timetable for that station

Setting Departure Times Manually

Click Setup departure Times



Then click set departure times



If we set departure times for stations along a route, then the train will not depart from the station before the departure time.

Train Route: Siding One FY_to_UPLinePlatform						
Station / Action	Journey Ti...	Wait Time	Duration (...)	Duration (f...	Departure ...	Delete Row
Siding One FY				0	00:00	☐
CornerUp	0	3	0	0	00:00	☐
UPLinePlatform	0	3	0	0	00:00	☐

Note that we are running at the default fast clock 1:10.

Note that if actions such as bell.py are present in the route, no departure times or durations are indicated for these.

The duration is defined as the time the train takes to travel between the previous station and the current station plus the wait time in the station. It is displayed in real seconds and fast minutes. The departure time is calculated automatically from the durations (and vice-versa).

We can either enter the journey times in real seconds, or the durations in fast minutes

We could now run a train over the route, measure the time taken, and enter the durations in fast minutes. Alternatively, you can note the time the train arrives at the station, and add a bit to allow the train to wait.

For example we enter 3 and 5 for the durations in fast minutes and toggle journey/wait times to get:

Train Route: Siding One FY_to_UPLinePlatform				
Station / Action	Duration (secs)	Duration (f mins)	Departure Time	Delete Row
Siding One FY		0	00:00	☐
CornerUp	18	3	00:03	☐
UPLinePlatform	30	5	00:08	☐

Calculates to the precision of 1 fast minute. 6 seconds = 1 fast minute

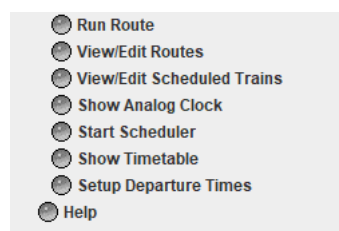
The departure times or durations update automatically.

We note that if we enter durations in seconds they are rounded down to the lowest 6 seconds

If you experiment you see that you can toggle the journey/wait times once more and enter the journey times in seconds

Setting Departure Times Automatically

To make it easier to obtain the durations the train can be run over the route, and the durations recoded automatically. The wait time in all or individual stations can be set so that the duration can be calculated (journey time + wait time) and stored.



To do this press Setup Departure Times. You get the options:

set departure times

Set Wait Time in Stations

This sets the default wait time in all stations. (in real secs). It can be overwritten in set departure times for a particular route.

wait time at station

set wait time at all stations

Note that the wait time will be rounded in practice to the nearest fast minute. At 10 times normal speed 1 fast minute will equal 6 seconds.

Run Train on Route to Set Journey Times

?

set departure times

set wait time in stations

run train on route to set journey times

set departure times

Pressing this option allows you to select a route to run. (An engine needs to be at the start of the route, and the engine need to be assigned using the setup button before running this routine).

☐ Express Train (no stopping)

☐ Setup Train in Section

☐ Enable Announcements

Pressing “Run Train on Route to Set Journey Times” brings up something like the following dialog:

Run Route and Record Journey Times

DownLinePlatform_to_Siding One FY

SP-MLU_to_DownLinePlatform

Siding One FY_to_UPLinePlatform

UPLinePlatform_to_SP-MLU

Cancel

Run Route

Select Siding One FY_to_UplinePlatform

Train Route: Siding One FY_to_UPLinePlatform

Station / Action	Journey Ti...	Wait Time	Duration (...)	Duration (f...	Departure ...	Delete Row
Siding One FY				0	00:00	☐
CornerUp	0	3	18	3	00:03	☐
UPLinePlatform	0	3	30	5	00:08	☐

Close

Toggle Journey/Wait Times

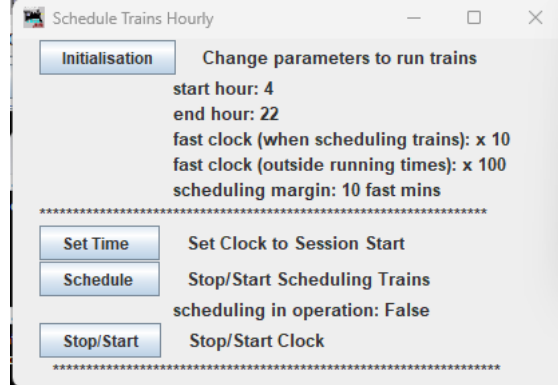
Set Wait Times

Run Train along route Siding One FY_to_UPLinePlatform now

Check train in required station
Then click to run route

Click Check train in required station, ensuring first that a engine is setup in (in this case Siding One FY).

The train will run over the route and record the time taken on each leg of the journey



The analogue clock also appears set to 00:00.

Initialisation: Change Running Period Parameters

Click “Initialisation”, A set of input boxes are displayed to enable setting of:

- The start of the running period
- The end of the running period
- The fast clock rate during the running period
- The fast clock rate outside the running period

Defaults are set and can be left as is for initial testing

Start the Scheduled Trains

- **Set the time to a period within the running period**

Click “Set Time”, The clock moves to 5 minutes before the start of the running period. It moves to the start of the running period at the fast clock rate for outside running times, then at the start of the running period, it changes to the running period fast rate.

- **Start the scheduling of the trains**

The trains will be scheduled according to the schedule table, and any timetables set up (see later) will be populated with the scheduled trains for the next 3 hours.

- **Start the clock**

The clock is initialised to stop when the start scheduling button is pressed. Click Stop/Start to restart the clock.

The trains will start scheduling

Stop Scheduling Trains

Click the Toggle button to stop scheduling trains and to restart scheduling

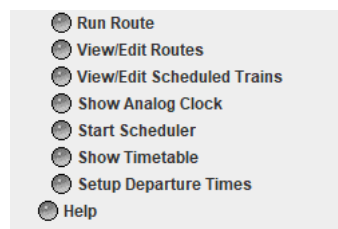
Viewing the Times of Scheduled Trains on Timetables

Introduction

A local timetable of any one station can be shown on the computer running JMRI. This is denoted as a local timetable.

Timetables can also be shown on remote tablets/computers. These are denoted MQTT timetables as they communicate with each other using MQTT.

Click the show timetable button.



A popup allowing the choice between a local timetable and MQTT timetables. MQTT timetables require some setting up. The local timetable works straight away.

?

timetables can be shown locally on this computer or on remote computers/tablets communicating by mqtt

scheduler needs to be on and clock running

set platforms and station groups

select station for local timetable

select station for mqtt timetable

Set Platforms and Station Groups

Press the ‘set platforms and station groups’ button and a table pops up showing all the stations on the system.

for a station group put the group name in one of the stations

—

□

×

Station	Platform	Station Group
CornerDown	Platform 1	Walden
CornerUp	Platform 2	
DownLinePlatform		
HeadShunt		
ML-MLD		
ML-MLU		
ParcelsSiding		
Scenic2Down		
Sector Plate		
Siding Lower FY		
Siding One FY		
Siding Three FY		
Siding Two FY		
Siding Upper FY		
SP-MLD		
SP-MLU		
SP-SD1		
SP-SD2		
SP-SD3		
SP-SPUR		
Traverser		
UPLinePlatform		

Close

You may find like the example above that the station names are rather cryptic, and not names that you would want to appear on a timetable. Looking at the diagram you see that CornerUp and Corner Down are two platforms at one place Walden. It would be nice to display the trains going in each direction on one timetable with the name Walden, and show what platform each train stops at. To do this the details for Walden had been filled in under station group, and the platform details under platform.

Timetables viewed on Computer running JMRI

Select the button “select station(s) for timetable. The following pops up:

select station(s) for timetable

×

CornerDown

CornerUp

DownLinePlatform

SP-MLU

Siding One FY

UPLinePlatform

Cancel

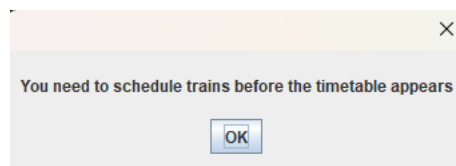
Show Timetable

Select the required station(s) that you want to display the timetable for. Note that only stations at which a train stops on a scheduled route are shown. (if a different station could be selected the displayed timetable would be blank).

We have selected two stations here by using the control button (using Windows). Similar for Linux and Mac.

A screen will be displayed reminding you that for timetables to be displayed the trains must be scheduled and the scheduler be running. If these conditions are satisfied you will get a timetable displayed

You may thus get a message



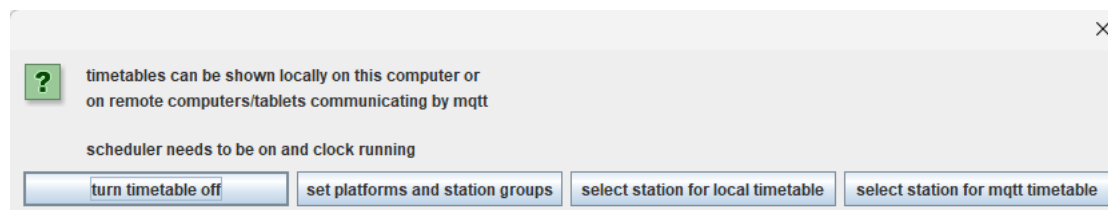
In which case you need to ensure that the clock is running and the train scheduler is running by clicking the start scheduler button.

A timetable similar to the below timetable will be shown:

Departures: Walden		04:00			
Arrival Time	Departure Time	Platform	From	Destination	Via
04:07	04:08	platform 2	Siding One FY	UPLinePlatform	-
04:46	04:48	platform 1	DownLinePlatform	Siding One FY	-
05:07	05:08	platform 2	Siding One FY	UPLinePlatform	-
05:46	05:48	platform 1	DownLinePlatform	Siding One FY	-
06:07	06:08	platform 2	Siding One FY	UPLinePlatform	-
06:46	06:48	platform 1	DownLinePlatform	Siding One FY	-

Turning Timetables Off

You will find that as the local timetable is refreshed every fast minute, it pops up above all other windows. If you wish to stop the timetable showing, click the 'turn timetable off' button.



Timetables viewed on Tablets / other Computers on same local Network

Introduction

Communication between the main computer and the tablets takes place using MQTT and Node Red,

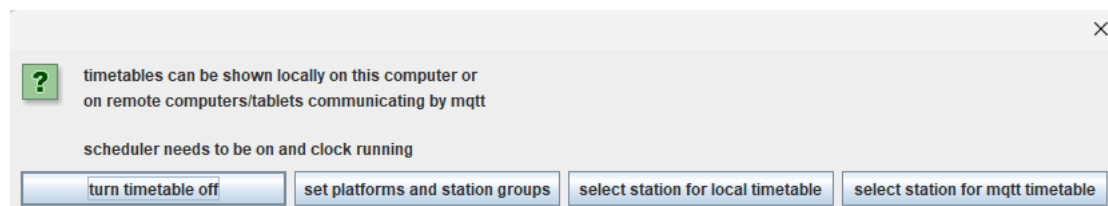
To display a timetable on a remote tablet or computer screen, we need to:

- 1) Set up the required MQTT environment using Mosquitto and Node Red on a local or remote computer
- 2) Set up a local or remote network connection
- 3) Setup a MQTT connection in JMRI
- 4) Generate the MQTT Node Red Code
- 5) Import the code to node red, and perform a trivial edit
- 6) Get the appropriate page on a browser to display the timetable

These timetables were inspired by a Rocrail MQTT implementation and modified from that implementation. The Rocrail implementation and hence the JMRI Node Red implementation is licenced as follows: [CC Attribution-Share Alike 4.0 International](#).

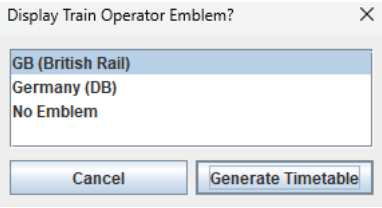
Generating the MQTT Node Red code

Click select station for MQTT timetable



Select the required station

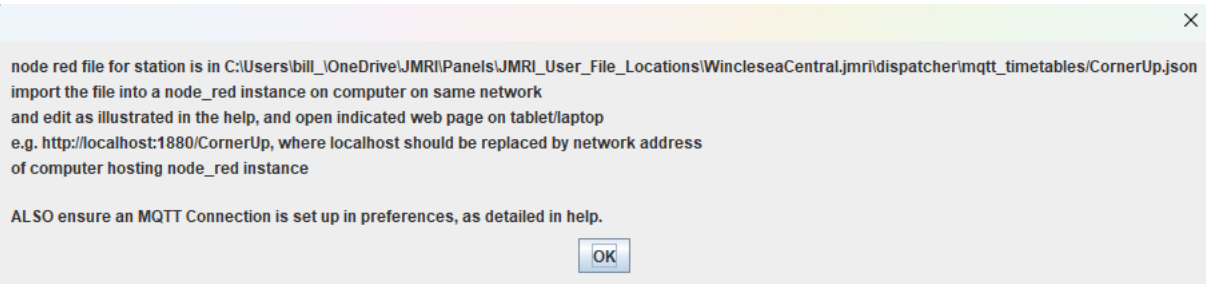
Proision is made to display an icon on the timetable. At present you can add a British Rail logo, a Deutsches Bahn logo or none.



Select the logo required and click Generate Timetable.

It is hoped that users will submit modifications to JMRI to add other emblems for different train operators.

Notification that a node red file has been produced in the user files directory

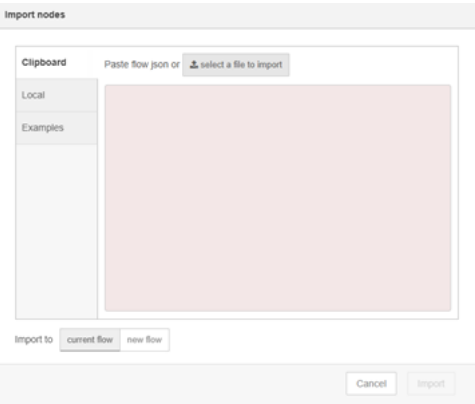
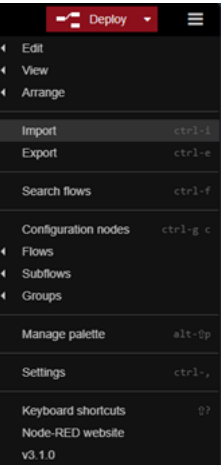


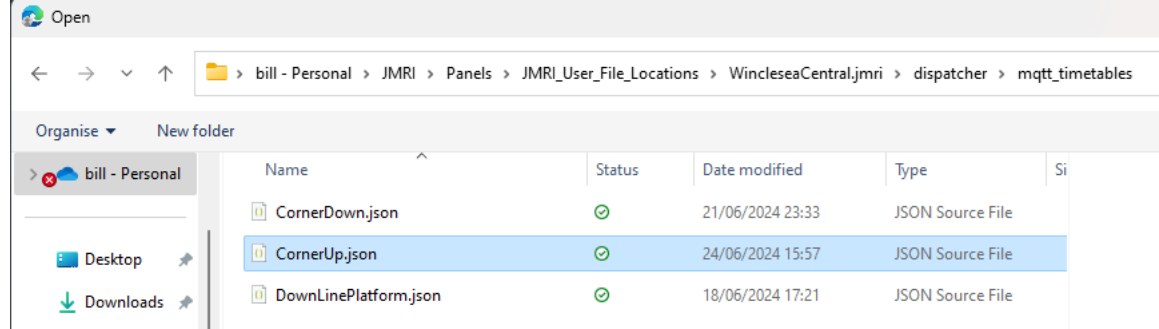
Setting up the required Node Red environment

Set up a MQTT broker such as mosquito on the same machine as is running JMRI, or one that is connected on the same network. You may use the instructions given in [Eclipse Mosquitto](#) if desired.

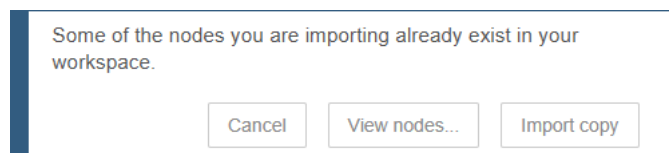
Set up Node Red as given in [Node-RED \(nodered.org\)](#). Note that if you are using a raspberry pi, with some installations Node Red comes installed. Note that there are many tutorials on this site which give you a good introduction to Node-red.

Start Node Red in a command file and start node-red in a browser using <http://localhost:1880> or similar. Import the node-red file generated in the previous section

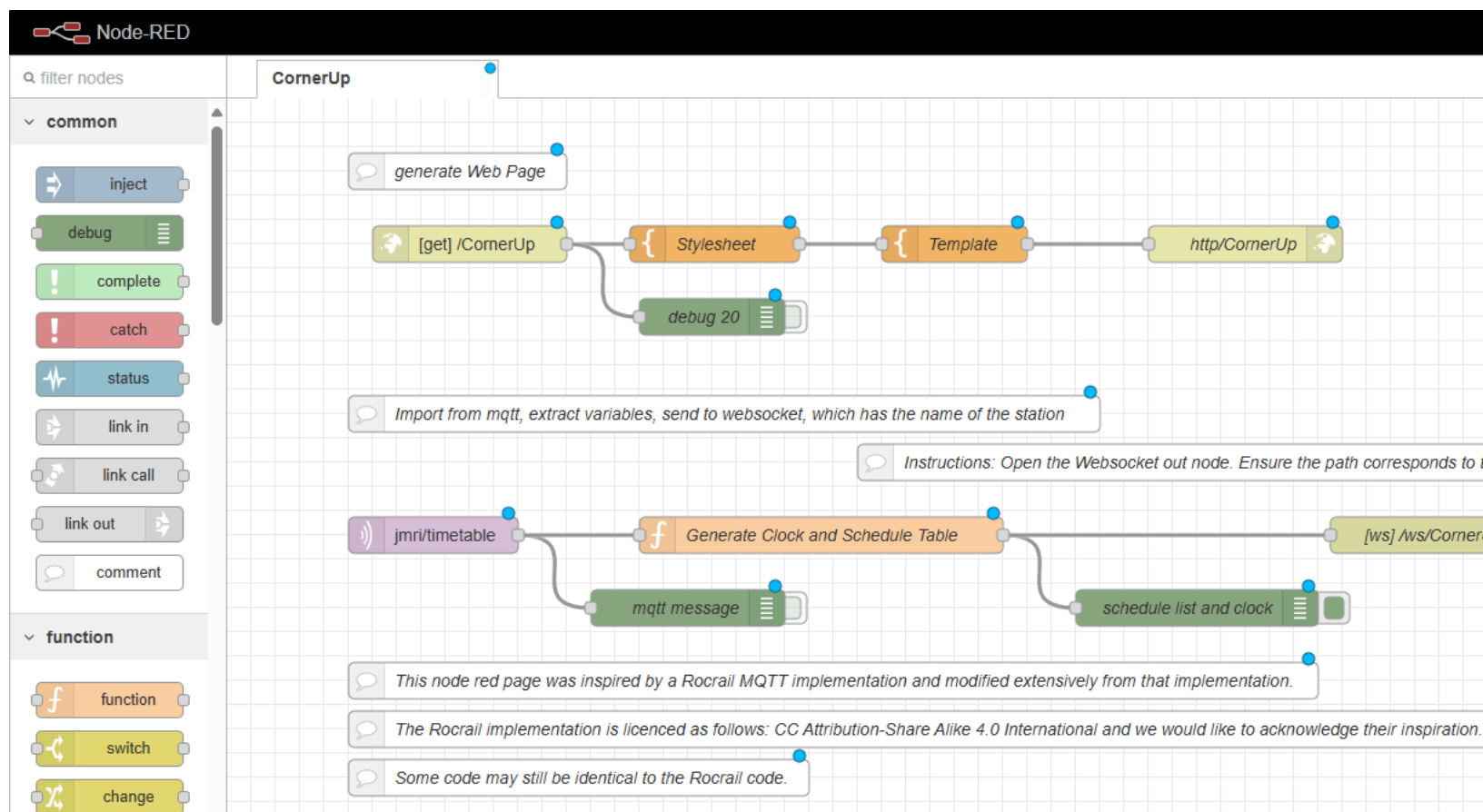




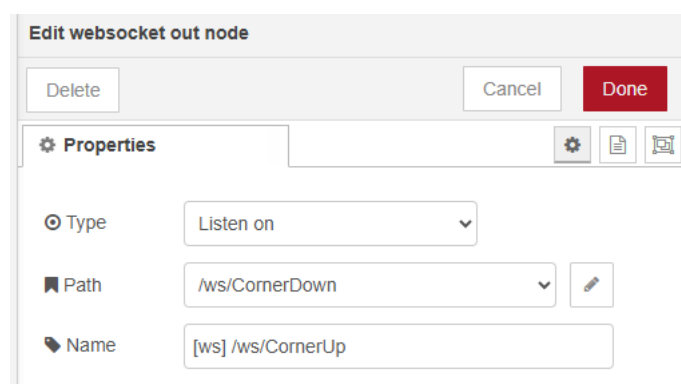
If you get the message



Click import copy



Open the web socket node as detailed in the comment. In this case it is the node labelled [ws] /ws/CornerUp. Double click that node, and you will get something like:



The path will not have been set to the correct value. Click the pencil to the right of the path, and enter the correct value /ws/CornerUp in our case.

Edit websocket out node > Edit websocket-listener node

Delete Cancel Update

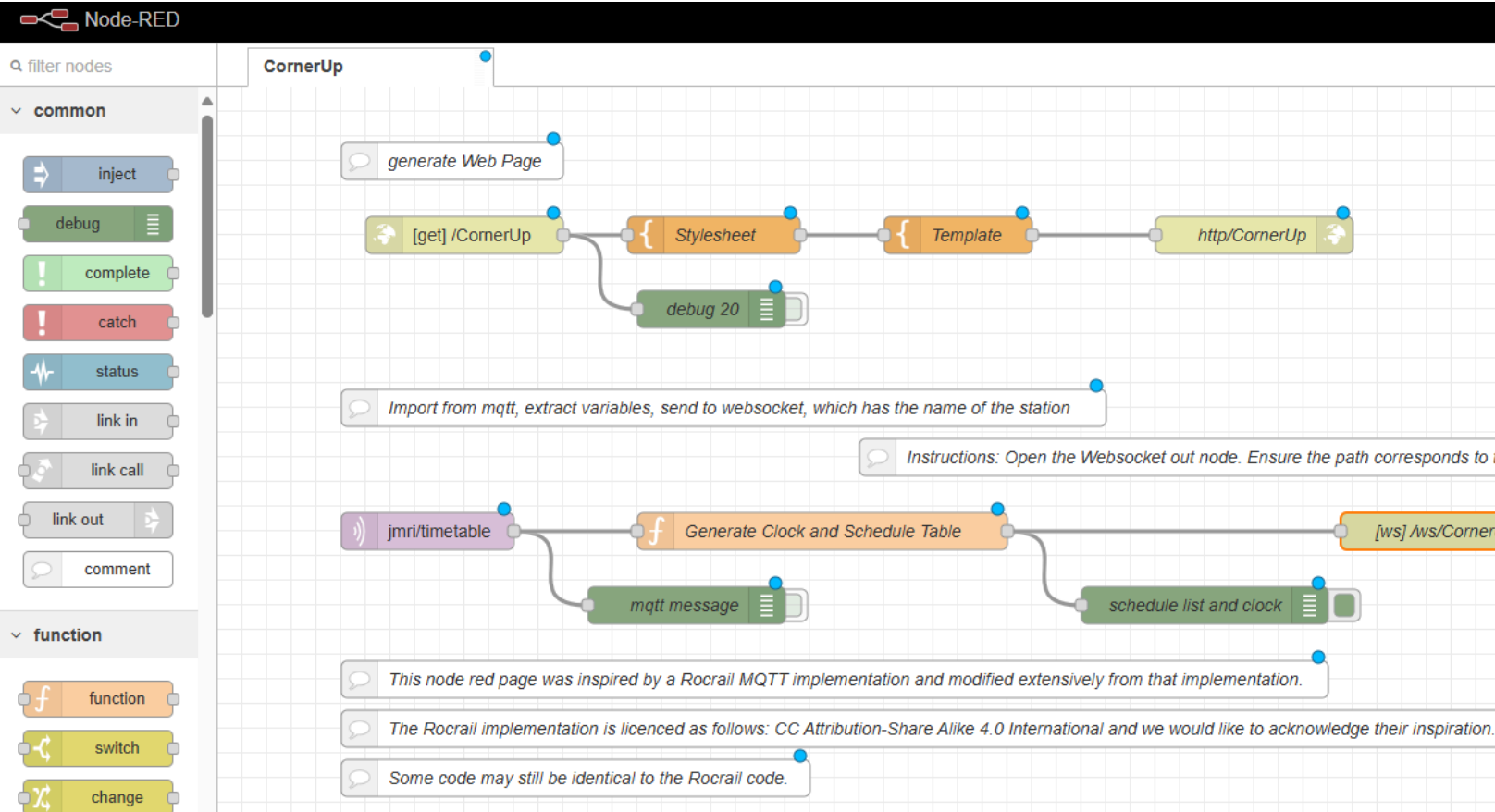
Properties

Path

Send/Receive

By default, **payload** will contain the data to be sent over, or received from a websocket. The listener can be configured to send or receive the entire message object as a JSON formatted string.

Click Update Done and Deploy



The blue dots will dissappear.

Setup a MQTT Connection in JMRI

A MQTT connection needs to be set up in JMRI to connect to the MQTT broker. Follow the instructions in the JMRI help. You will get something like:

Preferences

Window Help

Connections

MQTT MERG +

System manufacturer

MQTT

System connection

MQTT Connection

Settings

IP Address/Host Name: localhost

Connection Prefix: M2

Connection Name: MQTT

☐ Additional Connection Settings

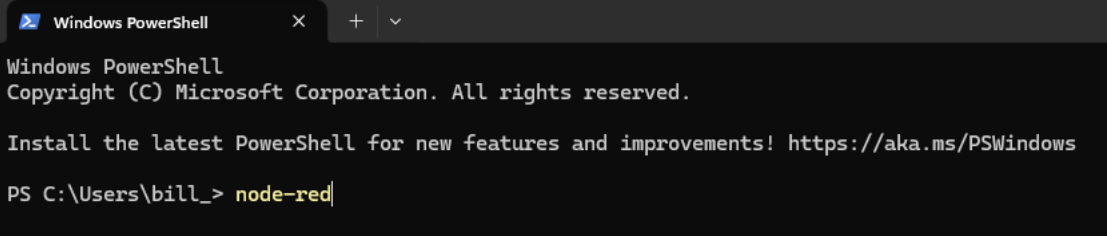
☐ Disable this Connection

Save

Display the MQTT timetable

Display the MQTT timetable by going to a browser and displaying <http://localhost:1880/StationName>. For example if the station name is Siding One FY, then you enter <http://localhost:1880/Siding One FY>

If the timetable does not appear ensure node-red is started, for example in windows



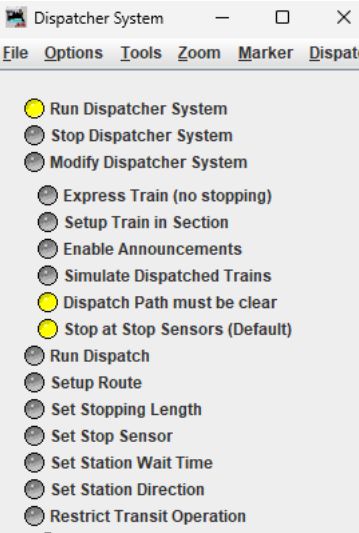
You should get something like:

Departures: Siding One FY					04:11	
Arrival	Departure	Train	From	Destination	Via	
04:26		Train4	DownLinePlatform	Siding One FY	Terminates Here	
	04:32	Train1	Siding One FY	UPLinePlatform	CornerUp	
04:56		Train4	DownLinePlatform	Siding One FY	Terminates Here	
	05:02	Train1	Siding One FY	UPLinePlatform	CornerUp	
05:26		Train4	DownLinePlatform	Siding One FY	Terminates Here	
	05:32	Train1	Siding One FY	UPLinePlatform	CornerUp	
05:56		Train4	DownLinePlatform	Siding One FY	Terminates Here	
	06:02	Train1	Siding One FY	UPLinePlatform	CornerUp	
06:26		Train4	DownLinePlatform	Siding One FY	Terminates Here	
	06:32	Train1	Siding One FY	UPLinePlatform	CornerUp	
06:56		Train4	DownLinePlatform	Siding One FY	Terminates Here	

Modifying the Behaviour of Running Trains

Provision is made to:

- Change the position where the trains stop, (using a stopping sensor or vary the calculated stopping length)
- Change the wait time in each station
- Set track to one way operation
- Stop transits starting for certain conditions. This will allow you to ensure that trains do not get trapped by other trains

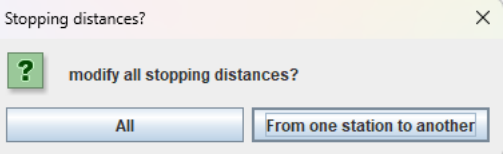


The buttons Set Stopping Length through to Restrict Transit Operation are used.

Set Stopping Distance

When a train stops at a station the stopping distance is calculated using the lengths of the blocks in the last section. If the train stops short or overruns, this distance can be changed.

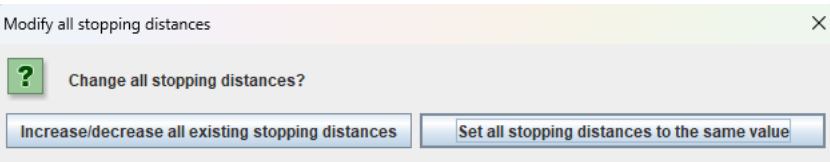
Click Set Stopping length.



The option is given to modify the stopping distances for all transits or just the stopping distance for one transit (from one station to another)

Modify All Stopping Distances

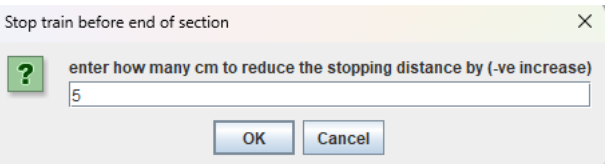
This modifies the stopping distance for all transits. Press All.



You may either

- increase or decrease existing stopping distances by the same amount, or
- delete any existing stopping distance changes, and increase or decrease existing stopping distances by the same amount

If we keep existing changes, but modify by a bit, press Increase/decrease all existing stopping distances



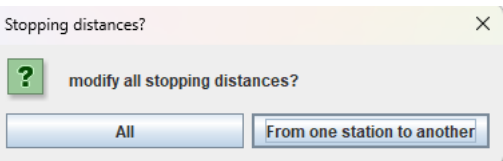
Enter the amount to decrease the stopping distance by



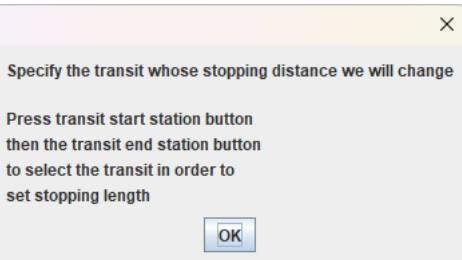
The train will now stop 5 cm short of the previous calculated position.

Setting all stopping distances to the same value is similar.

Modify Stopping Distance on one Transit



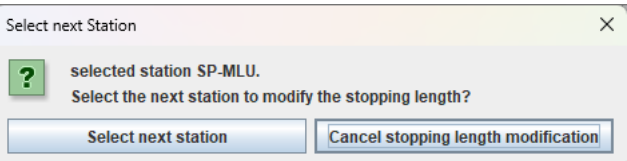
Click From One Station to Another.



You need to specify a transit going towards the station, so we select two stations, the first the start of the transit, the second the end.

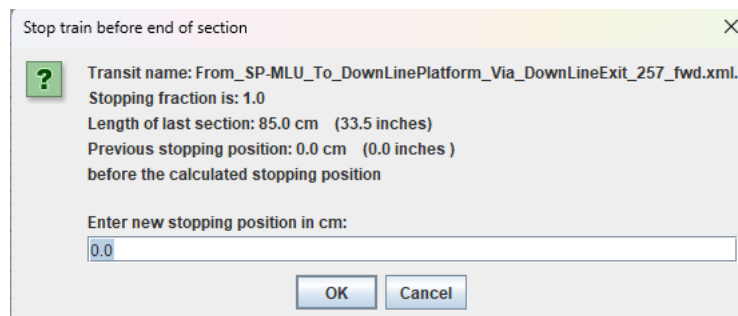
Suppose we want to change the stopping distance for transit going from SP_MLU to DownLinePlatform.

Select SP_MLU,





Click Modify Length

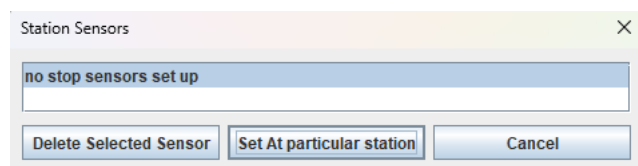


Specify the distance that the train will stop short of the calculated stopping distance

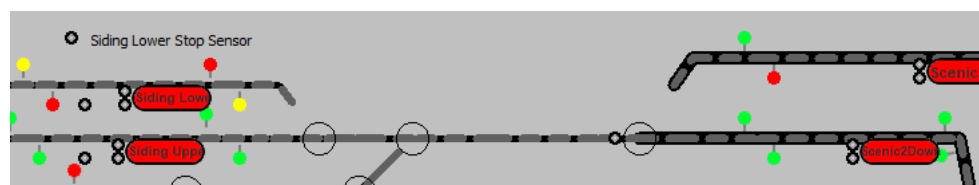
Set Stop Sensor



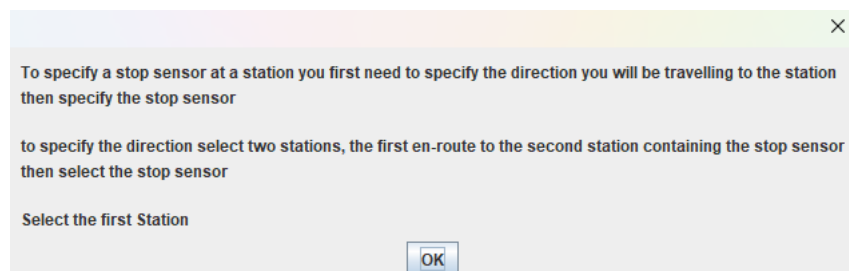
Click Set Stop Sensor



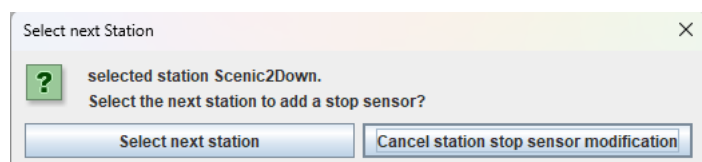
Suppose we wish to set stop sensor SidingLowerStopSensor at station Siding Lower FY, to operate when going in the direction Siding2Down to Siding Lower FY.



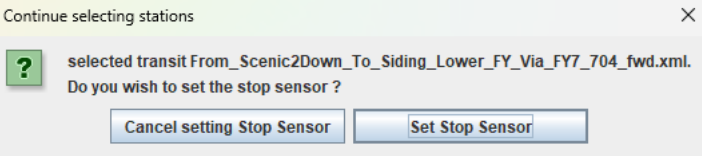
Click Set at particular station



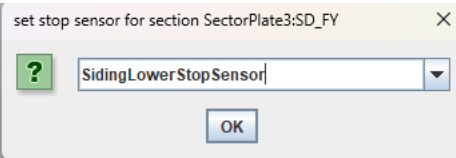
To specify the direction first click scenic2Down



Press 'select next station' then then the station containing the stop sensor, Siding Lower FY



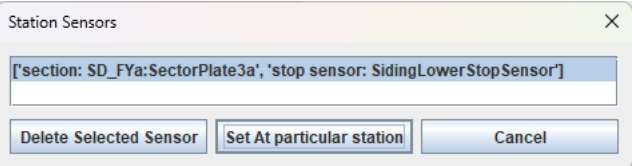
Click Set stop Sensor



Select the required stop sensor from the dropdown, in our case SidingLowerStopSensor and click OK

The set stop sensor routine finishes

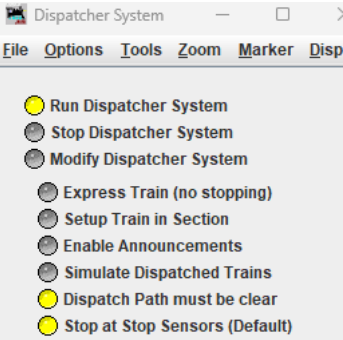
To check that you have set the sensor correctly select Set Stop Sensor again



It shows correctly that the section SD_FYa:SectorPlate3a has the stop sensor SidingLowerStopSensor.

Operation of Stop Sensors

All trains stopping at that section will stop at the stop sensor, provided the “Stop at Stop Sensors (Default)” button is enabled

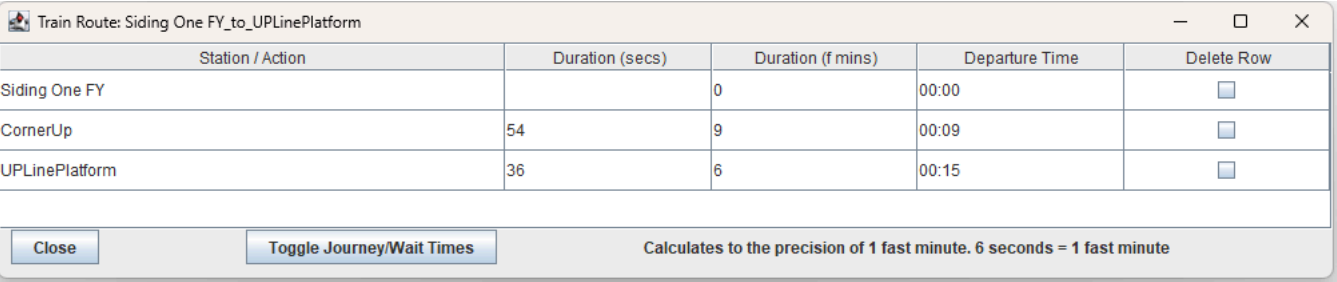


The default action for stop sensors may be overwritten by a route. If a stop sensor has been set at a station and a route stops at that station, the route can specify the stop sensor action.

The three possibilities are:

- Use Default
- Use Stop Sensor
- Use Speed Profile

Suppose one has created the route



If you then set a stop sensor at CornerUp and display the route again you get

Train Route: Siding One FY_to_UPLinePlatform

Station / Action	Duration (secs)	Duration (f mins)	Departure Time	Delete Row	Use of Stop Sensor
Siding One FY		0	00:00	☐	
CornerUp	54	9	00:09	☐	
UPLinePlatform	36	6	00:15	☐	

Close

Toggle Journey/Wait Times

Calculates to the precision of 1 fast minute. 6 seconds = 1 fast minute

The value of the stop sensor at CornerUp is not set and has the same effect as Use Default (e.g. use whatever the state of the Stop at Sensors (default) button).

If you wish to change the value, click the box and choose the required setting, which will override the default when running the route.

Train Route: Siding One FY_to_UPLinePlatform

Station / Action	Duration (secs)	Duration (f mins)	Departure Time	Delete Row	Use of Stop Sensor
Siding One FY		0	00:00	☐	
CornerUp	54	9	00:09	☐	Use Default
UPLinePlatform	36	6	00:15	☐	Use Default

Close

Toggle Journey/Wait Times

Calculates to the precision of 1 fast minute. 6 seconds = 1 fast minute

Use Stop Sensor

Use Speed Profile

Alternatively, when setting the route after the stop sensor has been set up, an extra option is provided at the station:

Continue selecting stations

?

selected station CornerUp.
Have you more stations on route?

Select another station

Complete Route

Set action to run in this station

Specify Stopping Mode at Station

If you don't specify the stopping mode, 'Use Default' will be set and shown when you view the route. If you set a stopping mode, that mode will be displayed on the route.

The action can be changed at any time by changing the option in the Use of Stop Sensor column. Only cells corresponding to stations with stop sensors will be able to be edited

Set Station Wait Time

The station wait times may be modified for all stations / stops, or at particular stations.

Click 'Set Station Wait Time'

- ☒ Run Dispatch

☐ Setup Route

☐ Set Stopping Length

☐ Set Stop Sensor

☐ Set Station Wait Time

☐ Set Station Direction

☐ Restrict Transit Operation

The following screen appears

Station Wait Times

?

modify all station wait times?

All

At particular station

Modify all station wait times

Click All

The following screen appears

Modify all station wait times

?

Change all station wait times?

Increase/decrease all existing station wait times

Set all station wait times to the same value

Either increase/decrease all wait times by same amount, or set all to same value

If setting to same value, click appropriate button, the following appears:

Pause train at beginning of section

?

enter how many secs to wait at station

5

OKCancel

Set the number of seconds to wait at station, and click OK

Pause train at beginning of section

?

new wait time: 5.0 secs

OKredo

Click OK

Modify the wait time at a particular station approach

Click 'At particular station'. The following screen appears:

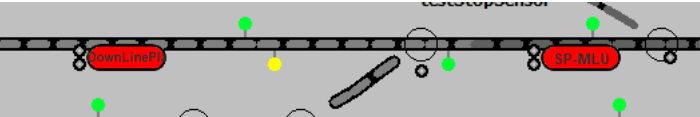
Press station buttons to select a transit (an approach to a station)

The wait time will be set for any train approaching in that direction (using any transit)

the wait time will be set at the section of the second station pressed

OK

Click a station on the approach to the station whose wait time we want to set.



Suppose we wish to set the wait time at station Downline Platform from direction SP-MLU. Click SP-MLU

Select next Station

?

selected station SP-MLU.
Select the next station to modify the station wait time?

Select next stationCancel station wait time modification

Select station DownLinePlatform

Continue selecting stations

?

selected station DownLinePlatform.
Do you wish to modify the station wait time ?

Cancel Station Wait Time modificationModify station wait time

Click Modify Station Wait Time

Pause train at beginning of section

?

Transit name: From_SP-MLU_To_DownLinePlatform_Via_DownLineExit_257_fwd.xml.
Station Wait Time is: 5.0

Enter new station wait time in secs:

5.0

OKCancel

The value is 5 (from the previous action of setting all wait times), set it to 3 say

new Station Wait Time: 3.0

OK

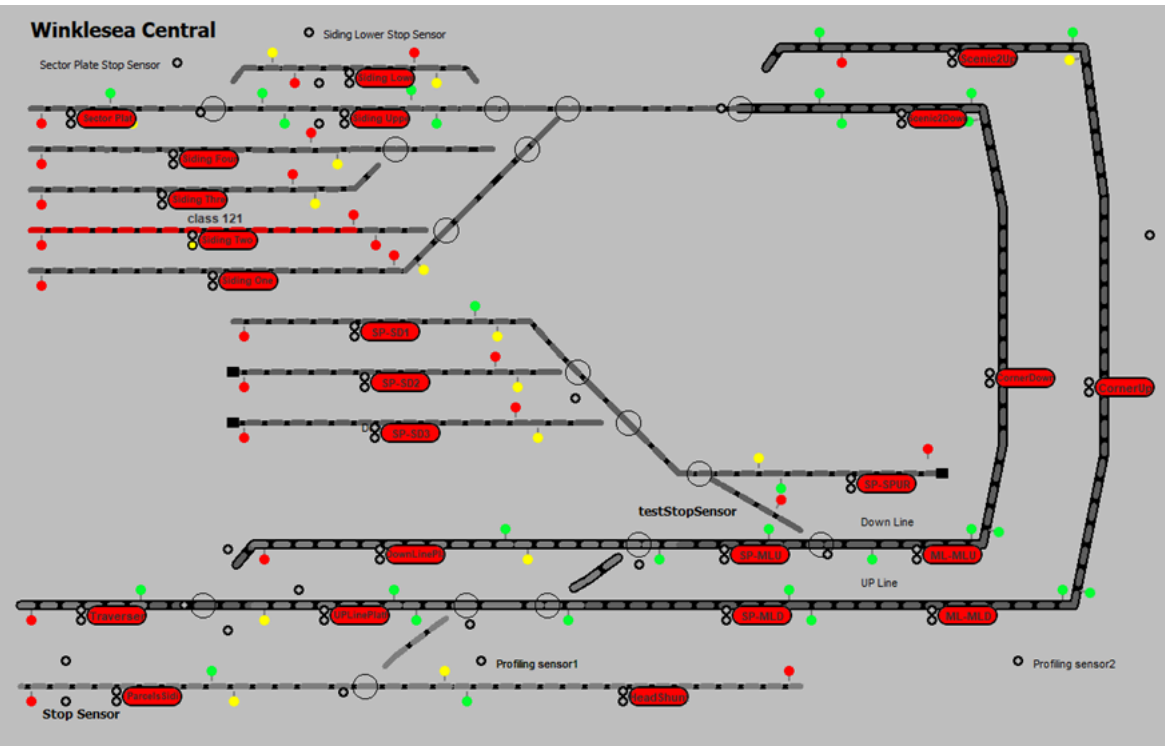
Set Station Direction

Setting Station Direction at a Station

The setting of station direction is best illustrated by an example.

On the following UK layout we wish to go from Siding One to UplinePlatform Via CornerUp, because we travel on the left, but the shortest route is via CornerDown. To make the train travel via Corner Up we have to make travel at CornerDown station one way in the direction of Siding One.

If you are happier you could enforce one-way working at both CornerUp and CornerDown, but the above argument shows that we only need to do so at the leg with the shorter length.



Setting the direction at CornerDown.

Click Set Station Direction

Station Directions

no inhibited directions

Reset direction restrictions

Set At particular station

Set At block

Click Set at particular station

Press station buttons to select a section in order to set one way working at that station
secect the second station adjacent to the first to indicate the direction

OK

Click CornerDown

Select next Station

?

selected station CornerDown.
Select the next station to modify the station direction?

Select next station

Cancel station wait time modification

Select Scenic2Down

Continue selecting stations

?

selected station Scenic2Down.
Do you wish to modify the station direction ?

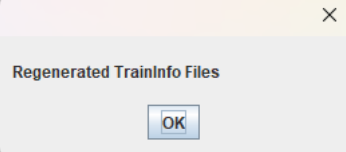
Allow only the direction from CornerDown towards Scenic2Down

Allow only the direction from Scenic2Down towards CornerDown

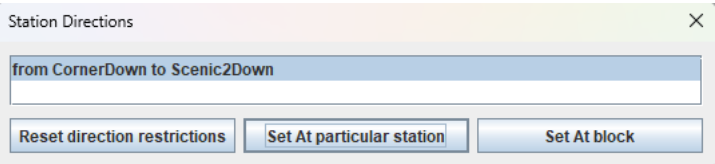
Allow 2-way working

Click Allow only the direction from CornerDown towards Scenic2Down.

The one way working is stored in the Train Info files. These are regenerated.



To confirm what we have done, Click Set the StationWaitTime Button again

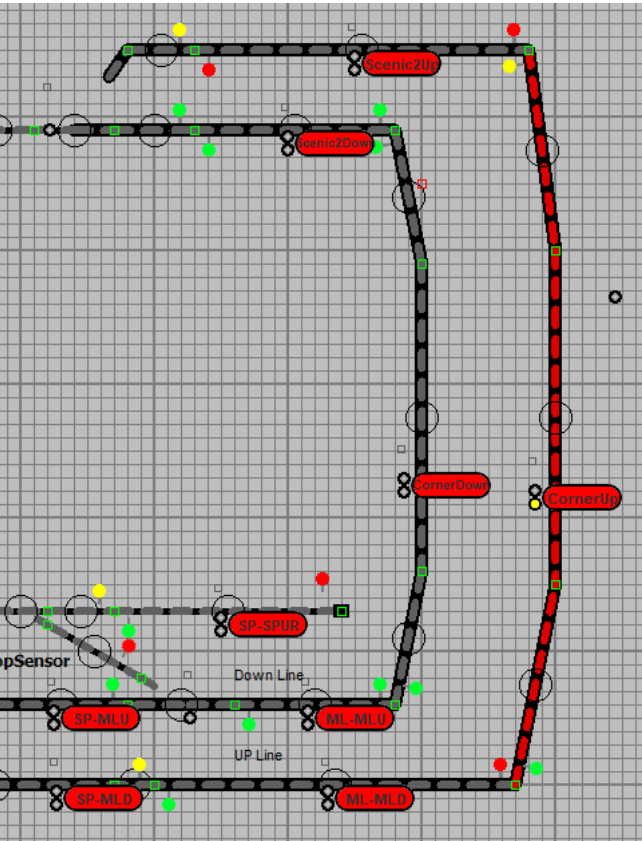


We see we have set one-way working in the direction CornerDown to Scenic2Down

Setting the Station Direction using a block

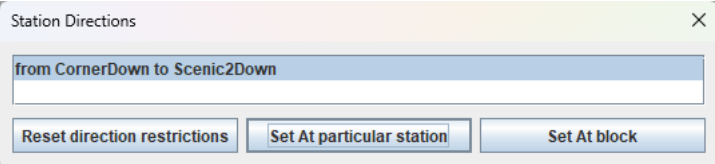
This is used where there is no station in the area required to be one way working, no possibility of putting one in (the area contains a turnout) without putting in extra blocks, or just no desire to have a station in the area.

The system of setting station direction using a block was designed for a system where the area of one way working had no possibility of putting a station in, but we will illustrate it by setting the block containing the station CornerUp to one way working, assuming that the stations Scenic2Up, CornerUp and ML-MLD do not exist.



The block CornerUp is highlighted.

Click Set Station Direction



Then click Set At Block,

Select block from list to
set one way working from that block
then select a block adjacent to the first to indicate the direction

OK

Click OK

choose first block

CornerDown

CornerUp

DownLineExit

DownLinePlatform

EntryTo Parcels

FY7

FY8

Head Shunt

ML-MLD

ML-MLU

Parcels Siding

SP-GAP

SP-MID

SP-MIDWAY

SP-MLD

SP-MLU

SP-P1

SP-P2

SP-P3

SP-P4

SP-SD1

SP-SD2

SP-SD3

SP-SPUR

Scenic2Down

Scenic2Up

Sector Plate

Sector Plate Dummy

Siding Four FY

Siding Lower FY

Siding One FY

Siding Three FY

Siding Two FY

Siding Upper FY

Traverser

Traverser-dummy

UPLineEntry

UPLinePlatform

OK

select block CornerUp, then click OK

The two neighbouring blocks are shown

choose neighbouring block

ML-MLD

Scenic2Up

OK

Select ML-MLD

Continue selecting stations

?

selected block CornerUp.
Do you wish to allow only one direction from this block ?

Allow only the direction from CornerUp towards ML-MLD

Allow only the direction from ML-MLD towards CornerUp

Allow 2-way working

Click Allow only the direction from CornerUp towards ML-MLD (UK trains run on the lefthand side.

The trainInfo Files are regenerated, and if one clicks Set the StationWaitTime Button again

- Run Dispatch
- Setup Route
- Set Stopping Length
- Set Stop Sensor
- Set Station Wait Time
- Set Station Direction
- Restrict Transit Operation

You get

Station Directions

from CornerDown to Scenic2Down
from CornerUp to ML-MLD

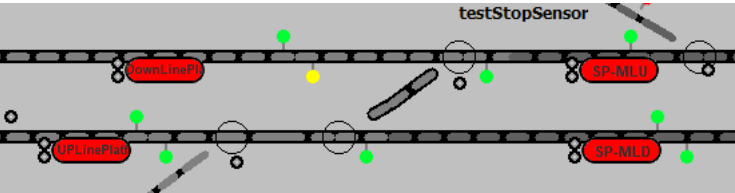
Reset direction restrictionsSet At particular stationSet At block

Click x to close.

Restrict Transit Operation

Introduction

This operation is best described by an example. Consider the following layout:



DownLinePlatform and UpLinePlatform are terminus stations (with a traverser at the end hence the continuations to the left). There is no easy way to set up a Traverser at the moment under Dispatcher, though it can be done with points and some Logix),

Consider the situation where trains are in Down and UpLinePlatforms. We are in the UK where trains travel on the left, and the train in UplinePlatform wants to move to DownlinePlatform, where it will pick up passengers before travelling in SP-MLU direction. To do this it will travel to SP-MLU and then to UplinePlatform.

If there is a train in UplinePlatform, this train will be trapped by the train in SP-MLU. To avoid this we need a rule, do not go on the journey UplinePlatform – SP-MLU – DownLinePlatform if there is a train in DownLinePlatform.

Creating the required Rule

- Run Dispatch
- Setup Route
- Set Stopping Length
- Set Stop Sensor
- Set Station Wait Time
- Set Station Direction
- Restrict Transit Operation

Click Restrict Transit Operation

Restrict Transit by specifying blocks which have to be free before transit can run

no transit restrictions

Remove Transit RestrictionSet Transit RestrictionCancel

Click Set Transit Restriction

Press station buttons to select a section in order to select transit, then specify the block that must be free

OK

Click OK, Select the start and end of the transit.

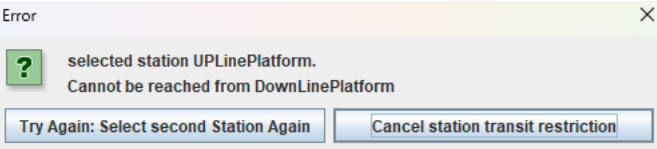
Click UpLinePlatform

Select next Station

? selected station ULinePlatform.
Select the next station to restrict operation of transit?

Select next stationCancel station transit restriction

Click Nest Station, and Click DownLinePlatform.



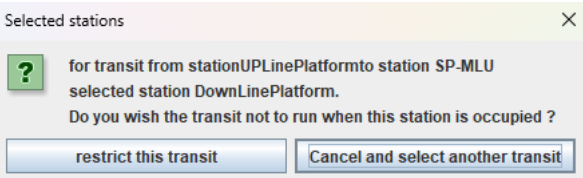
We get an error. Our reasoning is incorrect we must select SP-MLU, and apply a rule:

Do not go on the journey DownlinePlatform-SP-MLU if UplinePlatform is occupied

Select TryAgain, and Select SP-MLU

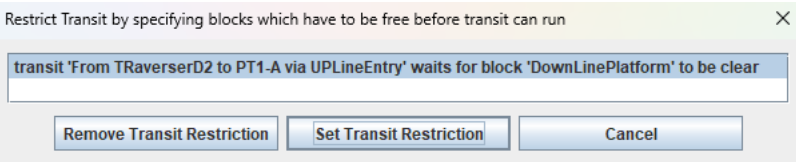
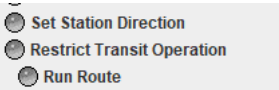


Press Select Station to Select its Block. Select Station UpLinePlatform



Click Restrict This Transit

To see what we have done, click Restrict Transit Operation again and we get:



If you have made a mistake, remove the transit restriction, otherwise Click Cancel

Keeping a Scheduled System Running

Inevitably when running trains using a scheduled system a fault will occur such as a train being derailed. It is not enough to place the train in the correct position as the system has knowledge about the direction of the train and its transit which may be disrupted. A simple tool is provided to mitigate these errors.

If the derailed train is dragged to the end of the transit, the transit may stop, but usually the train will go off in the wrong direction. If the train is removed and placed down in at the start of the next transit, the current transit may not complete.

Suppose we are running a schedule with a route which includes a transit for train J94 Saddle Tank Amazon and the train gets derailed in block FY7 (fiddle yard 7) which is the large red highlighted block.

 Modify System

Train	Delete Train	Block	Direction	Change Dir	Active Train	Transit	Del Transit
J94 Saddle Tank Amazon	☐	Siding One FY		☐			☐

 Modify System

Train	Delete Train	Block	Direction	Change Dir	Active Train	Transit	Del Transit
J94 Saddle Tank Amazon	☐	FY7	forward	☐	J94 Saddle Tank Amazon	From MD-1 to FY1_buffer via Scenic2D...	☐

Suppose we lift the train and transfer it to Siding One, we press the 'Refresh' button but the transit still shows. We click the check box by 'Del Transit' and the transit is closed down. The route disappears as well. We are now ready for the next route in the Schedule going from Siding One.

Suppose the next route on the schedule takes place moving from Siding One, and you notice the train is moving towards the buffers instead of away. Click the 'Change Dir' check box. The train changes the physical direction and changes the stored value of the direction for the train. The train will then behave correctly when the next transit in the sequence is called.

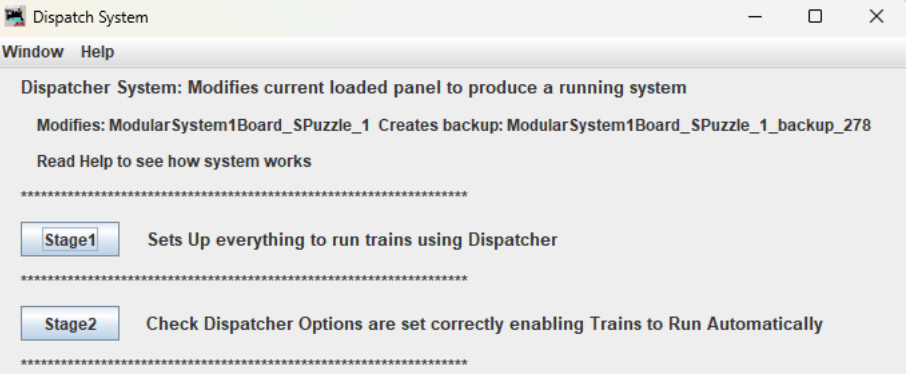
To add a new station or track

Edit the panel and add instructions to add any new stations:

- the new stations must be entered in the block table as described in a previous section
- open the new panel in panel editor, and change the panel to add new track and/or signals as required

Run the dispatcher system

- open the new panel, and re-run dispatcher system. If you have created a version of Dispatcher system before, and the trains were running OK, then you do not have to run stage 2, just Stage 1
- All the previous stations and icons will be removed, and new ones added automatically



Save the new panel, and restart JMRI.